

Contents

Topic 1: Consulting History	2
Topic 2: Why Consultants Are Needed and Their Types	5
Topic 3: Code Of Ethics	7
Topic 4: Business and IT Consulting	9
Topic 5: Consulting Process – Start to End	11
Topic 6: Consulting Engagements	15
Topic 7: Introduction to Business Analysis (BA).....	18
Topic 8: Fundamentals of Domain-Driven Design (DDD).....	24
Topic 9: Introduction to Domain-Driven Design (DDD)	28
Topic 10: Introducing the Enterprise Architecture	33
Topic 11: Enterprise Architecture (EA) Practices.....	37
Topic 12: An Introduction to Governance	50
Topic 13: Components and Facilitators of Corporate Governance	54
Topic 14 Part B: Ethics in Procurement	57
Flipboards Notes: (Not clear)	66

Topic 1: Consulting History

Origins and Evolution of the Consulting Profession

Key Themes:

1. Understanding the historical development of management consulting and their origin.
2. Recognizing consulting as a dynamic, ever-changing profession.
3. Evaluating consulting's role in shaping modern organizations.

Management Consultants and Consultancy in Modern Organizations

What Are Management Consultants? Management consultants are professionals who provide expert advice to organizations to improve their performance, solve complex problems, and enhance efficiency.

What Is Management Consultancy? Management consultancy is the professional service provided by management consultants. It includes systematic methods and tools to address organizational challenges.

Significant Role in Modern Organizations

Management consultants and consultancy play critical roles due to the following reasons:

1. **Dynamic Nature of Business:**
 - The business environment is constantly evolving with new technologies, regulations, and market trends.
2. **Focus on Improvement and Change:**
 - Consultants often act as **promoters of change**, helping organizations restructure or adapt to challenges like digital transformation, globalization, or competitive pressures.

Consulting Work: Dynamic, Ever-Shifting, and Contested

Dynamic:

- New firms regularly enter the consulting industry, offering fresh perspectives and solutions.
- Techniques and methods are continuously evolving to remain relevant.

Ever-Shifting:

- What is deemed effective today may not hold tomorrow. Consultants constantly reassess and update their approaches to align with the latest practices.

Contested:

- Despite its widespread adoption, consulting is not without criticism:
 - The effectiveness of some techniques can be questioned.
 - Critics argue that consultants may sometimes push their solutions to **keep demand going**.
- 1. **Consultants as Ubiquitous (Found Everywhere) Professionals:**
 - Consultants are involved in nearly every industry, from healthcare to tech to government.
- 2. **Consulting as Economic Activity:**
 - This involves viewing consultancy as a business service where advice and expertise are traded for fees.
- 3. **Human Element in Consulting:**
 - Emphasizes interpersonal skills like communication, relationship management, and the ability to work with diverse teams.
- 4. **Promoters of Change:**
 - Consultants drive innovation by introducing new methods or technologies and guiding organizations through transitions.
- 5. **Early Research on Consulting:**
 - Initial studies on consulting focused on its origins (e.g., Frederick Taylor's contributions) and its impact as a profession. **Business Press** raised awareness about consulting, playing a dual role by promoting consultants by sharing success stories while occasionally critiquing them. These critiques often included tips for clients on maximizing the value of consulting services. **The consultancies themselves** highlighted their own achievements, and **scholars** studied its deeper implications and history.

1. The Origins of Consulting

Two myths that obscured (distorted) the Engineering origins of Management Consulting:

- The origins of management consulting are often misunderstood due to two myths. First, **Frederick W. Taylor**, often called the "**father of scientific management**," is credited with laying the foundation for consulting by introducing systematic methods like time-and-motion studies. However, his role has been overstated to a mythical level, as his direct influence on the consulting industry was limited.
- Second, early "**efficiency experts**," who focused on improving operational workflows on factory floors, are often dismissed as not being true consultants. This perspective incorrectly narrows consulting's origins to strategic advisors for **top managers**, ignoring the significant contributions of these experts in optimizing work processes and driving efficiency.
- The limited diffusion of Taylor's ideas was addressed by dedicated management consulting firms that emerged in the early 1900s. These firms, such as Emerson Engineers and Charles Bedaux's consultancy, acted as "**carrying organizations in the early decades of the twentieth century**," adapting and spreading scientific management principles across various industries. By introducing proprietary systems and new techniques, they extended Taylor's concepts beyond their original scope and laid the groundwork for the professional consulting industry.

The Early Efficiency Consultants

Early scientific management engineers worked on a case-by-case/ temporary (**ad hoc**) basis, often individually or in small teams, offering advice to improve efficiency in specific workplaces. One notable figure was Harrington Emerson, founder of the **Emerson Engineers**.

He became familiar with Frederick Taylor's scientific management ideas and those of other engineers through the **American Society of Mechanical Engineers**. However, Emerson created his own unique (proprietary) system to optimize work processes and introduced the concept of **payment-by-results**, where employees' earnings were tied to their performance.

One of Emerson's key projects involved **reorganizing the machine and locomotive repair shops of the Santa Fe Railroad**, significantly improving their operations. Over his career, Emerson Engineers provided consultancy services to nearly 200 companies, helping them enhance productivity and streamline their operations.

Charles Eugène Bedaux's Role (1916):

- Marked the **true beginning of consulting** as a structured **business activity**. Viewed scientific management as a **major money-making opportunity**.
- **The Bedaux Efficiency System:**
 - Created a unique system of work measurement and **payment-by-results**, where employee pay was tied to their performance.
 - Highlighted by **authors** for its impact on implementing and extending scientific management principles.
- **Global Expansion (1930s):**
 - By the late 1930s, Bedaux's firm operated **internationally, becoming one of the first truly global consultancies**. Regions covered included the United States, Western Europe, Scandinavia, Eastern Europe, Africa, and Australia.
 - Played a crucial role in popularizing scientific management and setting the stage for consulting as a global profession.

Role of Efficiency Engineers During WWII:

- They became highly in demand due to the need to:
 - **Rapidly increase output** to meet wartime production demands.
 - **Integrate unskilled workers**, including a large number of women, into the workforce effectively.
- **Post-War Period:**
 - After WWII, the trend continued as companies focused on:
 - **Boosting productivity** to meet growing consumer demand.
 - **Reducing labor costs**, often influenced by the government.
- **MEC (Management Engineering Corporation):**
 - Founded in Pittsburgh in 1934, it became one of the most **successful and influential consulting firms** in the post-war era. Its **key innovation** was the development of MTM (**Methods-Time-Measurement**), a groundbreaking **work study technique**.
 - MTM provided a standardized way to analyze tasks and measure the time required for efficient work processes, helping industries improve productivity.

Three Factors attributed to the growth of consulting (Institutional Change):

- **"Knowledge Economies"** of consultants outweigh their costs.
(Knowledge Economies: Systems where knowledge and expertise are the main drivers of productivity and value.)
- Consultants **perpetuate (sustain or continue)** the demand for their services.
- Consultants **"compete with and displace" (replace)** senior and middle managers inside organizations.
- **Criticism of these perspectives:**
 - They **under-explore (fail to sufficiently address)** how broader social changes **enabled the growth** of management consulting.

Five such institutional changes are worth considering, with a focus on US history:

1. **Increased Number and Complexity of Corporate Organizations**
 - Transition from small, single-product, owner-run businesses to large, multidivisional corporations (**M-form**) to one dominated by large, complex corporations. By 1979 almost 90 per cent of large organizations had adopted the M-form.
(**M-form**: multidivisional structure is an organizational structure dividing firms (companies) into semi-autonomous units managed centrally.)
 - **Complex systems** were required for coordination, control and decision-making
 - Growth of **mergers and diversification**, e.g., GE (General Electric Corporation) with 100+ divisions by 1955. By the mid-1960s, **mergers** were topping 2,000 per year
(Merger: is the combination of two or more companies into a single entity, often to achieve greater efficiency, market power, or diversification.)
 - **New managerial skills** were required as full-time salaried executives came to dominate large corporations.
2. **Military-Industrial Complex**
 - Enormous growth in the U.S. military during WWII, with budgets increasing dramatically. (to over twenty-four times its pre-war size from 1939 to 1945)
 - US military spending also experienced a sharp increase, rising from \$2.5 billion in 1939 to \$162 billion in 1944 (1967 dollars)

- **Consulting firms** were called to help reorganize military organizational units
 - **After the war had ended**, high-ranking military personnel took positions with consulting firms, government departments, and in industrial organizations, **creating strong ties between the military, consulting firms, and industries.**
 - Innovations like the **PERT (Program Evaluation Review Technique) chart** developed by Booz Allen spread from military to corporate use. The war thus not only served as an immediate stimulus for strategy consulting, but as an irreversible one.
3. **Corporatization of Non-Corporate Sectors** (adopting business-like practices in areas like government, healthcare, and education)
- Spread of corporate beliefs, values, and practices to areas previously dominated by other institutional logics like government and healthcare. (The most significant changes occurred in **government** and **healthcare.**) Corporatization aligned with Weber's rationalization (emphasis on efficiency and predictability).
 - Strategy consultants, as purveyors (suppliers) of rational management techniques, promoted logical, data-driven methods originally developed in the corporate sector. They benefited as other sectors began adopting corporate-style management practices.
 - Federal, state, and local governments increasingly hired consulting firms, these firms aimed to bring cultural change to government operations by introducing corporate-style management.
 - **Cultural Shifts Promoted by Consultants:** Adoption of a private business mindset in non-corporate sectors, Emphasis on enhancing efficiency (achieving more with fewer resources), Introduction of measurable performance criteria (setting clear, quantifiable goals to assess success).
4. **Business Education**
- Business degrees rose significantly (Overall, from 3 per cent of all degrees in 1919 to 13 per cent in 1958), producing graduates who bolstered the consulting industry)
 - Business schools had a **symbiotic relationship** (mutually beneficial connection) with the management consulting industry, especially strategy consulting firms, in three key ways:
 1. **Producing Qualified Graduates:** Business schools create graduates who claim qualifications as management consultants. (However, these qualifications are widely accepted without strong evidence of their technical expertise for consulting. the actual technical qualifications to practice management **are largely unsubstantiated**) (**Largely unsubstantiated:** not strongly supported by evidence or proof.)
 2. **Advancing Management Knowledge:** Business schools help develop, codify (organize into a formal system), and share (disseminate) managerial knowledge.
 3. **Imparting Legitimacy:** Due to their affiliation with respected universities, business schools lend credibility to the ideas of 'management knowledge' and 'management progress.'
5. **Business Press**
- Raised awareness of consulting through positive (laudatory) coverage and advice for maximizing its benefits. (There was also some criticism of management consulting in the press. However, this was typically accompanied by advice on 'how to get the most' out of consulting engagements and avoid pitfalls)
 - Helped create management trends, driving demand for consulting services.
 - The business press allows management gurus (experts), well-known (prominent) CEOs, and consultants to spread their recommendations.
 - Generated client anxiety by sharing competitive success stories, reinforcing consulting's necessity.

Additional Info from Quiz

1. **Frederick Taylor's Role:** Often exaggerated but pivotal in early consulting
2. **Scientific Management Today:** Still relevant but needs ethical considerations to avoid misuse.
3. **Soft skills:** Considered more important than technical skills and **NOT** something innate (natural).
4. **VUCA World:** In a world characterized by **Volatility, Uncertainty, Complexity, and Ambiguity (VUCA)**, consulting demand continues to grow.
5. **Mindsets of practitioners:** Mindset of a strategist, consultant, business analyst, project manager

Topic 2: Why Consultants Are Needed and Their Types

Key Themes:

1. Understanding why consultants are required despite existing organizational structures.
2. Exploring the definition of consulting and consultants.
3. Categorizing the different types of consulting and their focus areas.
4. Identifying qualities and skillsets that make a good consultant.
5. Introducing frameworks and techniques used in consulting.

Keywords:

- Problem complexity
 - Types of consulting
 - Consultant definition
 - Consulting frameworks
 - Skill models in consulting
-

Why Do We Need Consultants?

Main Drivers for Consulting Services:

- **Complexity of Problems:** Modern challenges in businesses arise from:
 - **1. Business Issues:** Problems related to a company's strategy or market activities.
 - A company might not know the best way to grow, enter a new market, or compete with others.
 - Example: Imagine a bakery wants to start selling online but doesn't know how to create an online store, attract customers, or handle deliveries.
 - **2. Organizational Issues:** Challenges within the structure or culture of a company.
 - These problems deal with how people in a company work together or how the company is set up.
 - Example: A company with departments that don't communicate well may find it hard to finish projects on time. It's like a sports team where players don't pass the ball to each other.
 - **3. Technological Challenges:** Issues arising from new technology or outdated systems.
 - Problems occur when technology doesn't work well, is hard to use, or needs upgrading.
 - Example: A school switching from paper records to an online system might struggle with setting it up or training teachers to use it.
 - **4. Industry-Specific Challenges:** Problems unique to the company's industry.
 - Some industries have special rules or challenges.
 - Example: A car company may face difficulties meeting new safety regulations. It's like students preparing for new exam rules that weren't there before.

Value Consultants Bring:

1. **Variety of Skills and Expertise:** Specialized knowledge across industries and domains.
2. **Objectivity:** Neutral perspective in assessing problems and solutions.
3. **Advice on External Environments:** Identifying trends, market analysis, and benchmarking.

Definitions:

Consulting:

- A structured process where professionals analyze a client's issue and propose documented recommendations to address it.

Consultant:

- A professionally trained individual in Consulting process who solves the client's problem through his recommendations.
 - A major emphasis (special importance or focus) is placed on the steps and methods a consultant follows to carefully analyze the problem and develop recommendations that are well-reasoned and effective.
-

Types of Consulting:

Consulting services can be broadly classified into the following categories:

1. **Strategy Consulting:** High-level business strategies like market entry or mergers (Explained in topic 1).
2. **Financial, Accounting, and Audit (Formal examination of financial records) Consulting:** Budgeting, tax planning, and auditing services.
3. **General Management Consulting:** Addressing organizational challenges and enhancing operations.
4. **Industry Operational Consulting:** Tailored solutions for specific industries (e.g., manufacturing, healthcare).
5. **Information Technology Consulting:** Focused on IT systems, infrastructure, and cybersecurity.
6. **Function-Focused Consulting:** Specializing in specific business functions like HR or marketing.
7. **Hybrid Consulting:** Combining expertise across multiple domains.

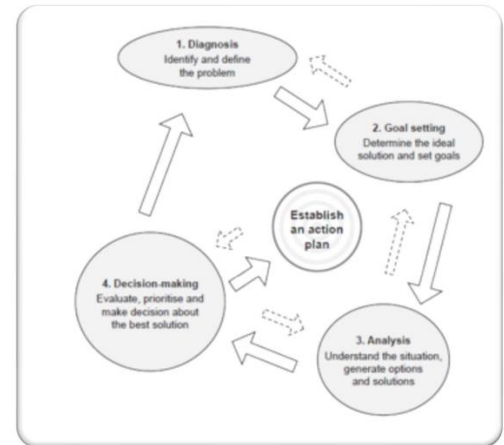
Who Can Be a Good Consultant?

Key Success Factors:

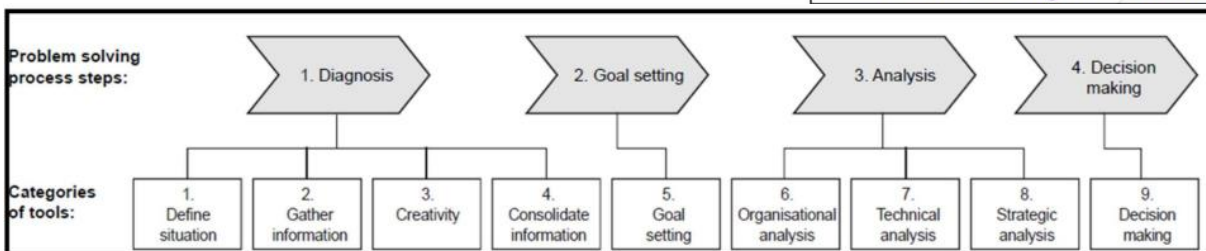
1. **Deep and Broad Expertise:** Knowledge across multiple industries and specializations.
2. **Quality-Focused Deliverables:** Providing actionable and impactful recommendations.
3. **Strong Relationships:** Building trust and maintaining long-term client engagements.
4. **Commitment to Continuous Learning:** Bridging personal knowledge gaps.

Frameworks and Techniques:

- Consultants use structured frameworks to systematically address client issues.
- Problem-solving frameworks typically involve:
 - **Problem Identification**
 - **Data Collection and Analysis**
 - **Solution Development**
 - **Implementation and Monitoring**



Problem solving framework



Problem solving process step	Type of thinking	Warning – Most common pitfalls ☹️
Diagnosis	Integrative 	Taking a narrow, functional view or being too broad to generate effective responses.
Goal setting	Visionary and pragmatic 	Only top-down or bottom-up thinking – no coordination and agreement between top-management and operational level on realistic objectives.
Analysis	Divergent 	Starting out looking for the ONE correct answer/solution. Shooting down solutions that appear logical.
Decision making	Convergent 	Allowing politics, ego and emotions rather than logic to decide the outcome. Continuing to be creative, without applying sufficient analysis and judgement.



Additional Insights from Quiz and Slides:

1. **Nature of Problems:** Consultants handle challenges that could be Resolving defects, Leveraging opportunities or mix of both
2. **Skill Models:** Consultants benefit from multi-dimensional skills like:
 - **T-Shaped Model:** **Horizontal bar:** Broad general knowledge, **Vertical bar:** Deep expertise in one area.
 - **Pi-Shaped Model:** **Two vertical bars:** Expertise in two areas. **Horizontal bar:** Broad general knowledge connecting them.
 - **V-Shaped Model:** Focuses less on broad general knowledge and more on **deep practical expertise** in two fields with a strong ability to implement solutions effectively.
3. **Learning Curve:** Most skill development happens **on-the-job** rather than through formal education.
4. **Iterative Problem Solving:** Consulting processes often involve repeating steps to refine solutions, rather than following a rigid linear sequence.

Topic 3: Code Of Ethics

Key Themes

1. **Understanding Ethical Standards in Consulting:** Ethical principles form the backbone of the consulting profession.
2. **Challenges and Ethical Dilemmas:** Recognizing and addressing ethical challenges ensures professionalism and trustworthiness in client relationships.

Key Words

- **Confidentiality:** Search when discussing information handling or disclosure.
- **Unrealistic Expectations:** Relevant for promises or outcome assurance.
- **Commissions and Financial Interests:** Focus on third-party benefits and financial transparency.
- **Conflicting Assignments:** Use for potential role conflicts or disclosures.
- **Professionalism:** Important for overall consultant conduct.
- **Ethical Challenges:** Refer to dilemmas, greed, and conflicts of values.
- **Scientific Management:** Search for historical influences and ethical implications.

Key Principles and Topics

1. **Confidentiality:**
 - Consultants must treat client information as strictly confidential.
 - Privileged information gained during assignments should not be exploited personally or shared with others.
2. **Unrealistic Expectations:**
 - Consultants must not overpromise or assure clients of guaranteed outcomes.
 - Ethical practice requires setting realistic and achievable expectations for consulting services.
3. **Commissions and Financial Interests:**
 - Acceptance of benefits, remuneration (bribe) or commissions from third parties connected to recommendations requires client knowledge and consent.
 - Disclosure of financial interests in any recommended goods or services is mandatory (required).
4. **Assignments Accepted:**
 - Consultants should only accept tasks for which they have the required or necessary skills and expertise.
5. **Conflicting Assignments:**
 - Acting in conflicting roles without informing all relevant parties (acting simultaneously) is unethical.
 - Advance disclosure of potential conflicts is necessary.
6. **Conferring With Clients:**
 - Before starting an assignment, mutual understanding regarding objectives, scope, work plan, and fee arrangements must be established.
 - Any personal or financial interests influencing the assignment must be disclosed.
7. **Objectivity of Advice:**
 - Consultants are obligated to provide independent and unbiased (objective) advice.
8. **Timeliness:**
 - Assignments should be completed according to agreed timelines and work plans.
9. **Recruiting:**
 - A consultant will refrain from inviting an employee of a client to consider alternative employment without prior discussion with the client.
10. **Professional Approach:**
 - Consultants must maintain professionalism in all dealings with clients, the public and fellow Consultants.
 - In support of this, Consultants must not:
 1. Engage in any activity, which might impair their judgment
 2. Make false or misleading statements
 3. Act in any way, which discredits the Consulting community

Ethical Challenges and Unethical Behaviors

1. **Nature of Unethical Behavior:**
 - Consultation is a relatively young profession and is still undergoing the long and arduous (extremely difficult) process of standardization and professionalization.

- Historical implications (i.e., suggesting something without explicitly stating it), such as those of scientific management, have at times dehumanized workers, leading to ethical compromises. For instance, Taylor's approach often treated human beings as machines or parts of machines. Workers were pictured as stupid and unable to perform tasks adequately without supervision.
 - Scientific management required managers to strictly enforce methods, often treating workers as tools for efficiency rather than individuals with value. This clashed with the **Kantian moral principle**, which emphasizes treating people as ends in themselves (respecting their rights and value as a human being), creating ethical pressure on efficiency experts.
- 2. Duty to Clients (An Unconditional Good?):**
- Ethical dilemmas arise when duties to client conflict with broader moral obligations (e.g., withholding truth from staff or supporting executives' unsavory/unpleasant decisions).
 - Consultants may face scenarios where their personal values clash with client expectations.
- 3. Greed A Fundamental Driver of Bad Behavior:**
- Consultants aren't greedier than anyone else, it is only that they are in a unique position to capitalize on it
 - Greed can lead consultants to prioritize revenue over client needs.
 - Consultants are typically called in to solve problems, that is when clients (or potential clients) are the most vulnerable.
 - A disturbing amount of material derived from work for competing companies will routinely turn up on work for another, even on work for competing companies in the same industry

Additional Quiz Info

- 1. Decision-Making:** To minimize bias, decisions should involve isolated preparation followed by group discussions for consensus (agreement).
- 2. Confidentiality:** Confidentiality of information is **not** an eternal restriction.
- 3. Recruitment Practices:** A client (**Not only the consultant**) will refrain from inviting an employee of a **consultant team** to consider alternative employment without prior discussion with the firm.
- 4. Technology's Role:** Technology is an enabler of business outcomes, not an end in itself.
- 5. Protecting information and knowledge is typical for:** the exploiting phase or track of business of an organization.

Exploratory Phase:

- **Focus:** Innovation and experimentation.
- **Goal:** Discover new opportunities, markets, or ideas.
- **Characteristics:**
 - Involves high uncertainty and risk.
 - Encourages creativity and trial-and-error approaches.
 - May include research, development of new products, or testing novel business models.
- **Example: A tech company investing in AI research to create a groundbreaking product.**

Exploiting Phase:

- **Focus:** Optimization and efficiency.
 - **Goal:** Maximize value from existing resources, products, or markets.
 - **Characteristics:**
 - Involves refining processes and scaling proven ideas.
 - Focuses on cost reduction, productivity, and predictable results.
 - Prioritizes stability over risk-taking.
 - **Example: The same tech company optimizing production and marketing for an established product to increase profitability.**
- Key Difference:**
- **Exploratory Phase:** About *creating new value* through innovation.
 - **Exploiting Phase:** About *extracting maximum value* from existing assets or knowledge.

Topic 4: Business and IT Consulting

Key Themes:

- **Increasing Strategic Role of IT**
- **Growing Symbiotic (mutual) Relationship Between Business Consulting and IT Consulting**
- **Profile of an Ideal Business and IT Consultant**

Keywords

- **ERP, CRM:** For IT's role in business processes.
- **Business-IT Integration:** For the symbiotic relationship discussion.
- **IT Governance, IT Security:** For detailed IT consulting services.
- **Attributes of Consultants:** For ideal consultant profiles.

1. Introduction: Business Converging (Combining) with IT

- **IT's Central Role:** Tools like ERP (Enterprise Resource Planning) and CRM (Customer Relationship Management) demonstrate that IT is essential for implementing strategic business changes.
- **IT is imperative (necessary)**
- **Emergence of Key IT Trends:** Business Analytics, Collaboration, Workflow, Artificial Intelligence (AI), and the convergence of the Internet and mobile telephony have made IT central to strategic changes.
- **Interdependence of Business and IT:** Business consulting cannot occur without IT. IT consulting must include a business component to move beyond just gathering requirements (requirements solicitation) and developing systems.
- **Thomas Carlyle's Eulogy (Tribute) for Tools:**
 - Carlyle emphasized the transformative power of tools in human progress.
 - Tools extend human capabilities and improve life.
 - His 200-year-old vision of tools' relevance is realized by a tool we call IT
 - Relevance in IT:
 - IT is the modern equivalent of those transformative tools.
 - IT systems like ERP and CRM are essential in driving business innovation and efficiency today.
- **Key Insight:** No business consulting can happen without IT, and vice versa. The two are intertwined to meet modern business challenges.

2. Definition of Business and IT Consulting

- **Business-IT Theme:**
 - **IT Strategy for Business:**
 - A plan that outlines how IT will be used to support and enhance business goals, ensuring that IT investments align with the company's strategic objectives.
 - **IT Planning, IT Application Portfolio Planning:**
 - **IT Planning:** The process of determining the IT needs of an organization, including infrastructure, resources, and technology investments.
 - **IT Application Portfolio Planning:** Managing and optimizing the collection of IT applications to align with business priorities and reduce redundancy or inefficiency.
 - **E-business Strategy:**
 - A strategy that leverages electronic means, particularly the internet, to conduct business processes, including sales, customer service, and internal operations.
 - **COTS-Enabled Business Process Design:**
 - Using **Commercial Off-The-Shelf (COTS)** software to design and implement business processes, focusing on integrating pre-built solutions to meet specific organizational needs.
 - **Cost Management: Integrated Supply Chain:**
 - Managing costs across the entire supply chain by integrating processes from suppliers to customers, aiming for efficiency and cost-effectiveness.
 - **Customer-Centric: CRM Process Redesign:**
 - Redesigning Customer Relationship Management (CRM) processes to focus on improving customer interactions and satisfaction by making the customer the central focus of business operations.
 - **IT-Initiated Organization Change:**
 - Organizational changes driven by IT innovations or implementations, such as adopting new software systems or digital transformation initiatives to improve business performance.
- **Focused IT Management Theme:**
 - **IT Architecture:**

- The structural design of an organization's IT systems, including hardware, software, networks, and databases, to ensure they align with business goals and support operations effectively.
- **IT Service Management:**
 - The process of managing IT services to meet the needs of the organization, focusing on delivering high-quality IT services efficiently and effectively.
- **IT Process Management, Simplification, and Optimization:**
 - Managing and improving IT processes to make them more streamlined and efficient, reducing complexity and enhancing performance.
- **IT Governance:**
 - A framework that ensures IT investments and processes align with business objectives and deliver value while managing risks and resources effectively.
- **IT Process Quality Improvement Consulting:**
 - Consulting services that focus on enhancing the quality of IT processes, ensuring they meet industry standards and best practices for efficiency and reliability.
- **IT Knowledge Management:**
 - The practice of capturing, organizing, and sharing IT knowledge within an organization to improve decision-making, efficiency, and innovation.
- **Business Continuity and Disaster Recovery Consulting:**
 - Consulting services that help organizations prepare for and recover from disruptions, ensuring business operations can continue or quickly resume after a disaster.
- **Planning, Budgeting, and Monitoring IT Programmes:**
 - The process of planning IT projects, allocating budgets, and monitoring their progress to ensure they are completed on time, within budget, and meet business requirements.
- **IT Security, Data Security, and Prevention of Fraud and Intrusion:**
 - Measures and strategies to protect IT systems and data from unauthorized access, cyber threats, fraud, and breaches, ensuring the integrity and confidentiality of information.

3. Business-IT Consulting: A Growing Symbiotic Relationship

- **Integrated Approach:** Business Consulting and IT Consulting must align closely to ensure effective implementation of strategies.
- **Complexity:** The increasing power and capability of IT add complexity, necessitating strong integration efforts.

4. The Ideal Business and IT Consultant

- **Attributes:**
 - Objectivity, Teamwork, Simple Language, Distancing from Client's Politics, Business Attire(formal or semi-formal clothing), Continuous learning, Ability to create a niche(The skill to identify and specialize in a specific area), Enthusiasm

5. Wrap-Up

- **Inseparable Relationship:** The growth in IT's role ensures that business and IT consulting are becoming inseparable for competitive advantage.

Engineering Terms

- **ERP (Enterprise Resource Planning):** Software systems that manage core business processes.
- **CRM (Customer Relationship Management):** Tools and strategies for managing a company's interactions with current and potential customers.
- **IT Governance:** Frameworks that ensure IT investments support business goals.
- **COTS (Commercial Off-The-Shelf):** Ready-made software solutions.

Topic 5: Consulting Process – Start to End

Key Themes:

- Understand the start-to-end consulting process and its sub-processes.
- Learn planning aspects for implementing consulting recommendations.
- Gain an overview of managing follow-on engagements.

Keywords:

- Consulting process steps.
- Value-enabling sub-processes.
- Walkthroughs.
- Scope changes

Key Concepts

Consulting Process Overview

- A series of logical steps aiming at maximizing the degree of success.
- Two main activities:
 1. **Selling Consulting Services:**
 - Some firms separate selling (business development) from delivery (engagement execution).
 2. **Delivering Recommendations.**

Head Intellectual Capital Success Factors: CPA and RCA (Additional Info from lecture)

Intellectual Capital: The collective knowledge, skills, and expertise of an organization or its people, considered a key asset for achieving success in consulting. It includes three main components:

1. **Human Capital:** Skills, creativity, and expertise of individuals.
2. **Structural Capital:** Tools, processes, and systems that support the work.
3. **Relational Capital:** Relationships with clients, partners, and stakeholders.

Success Factors: CPA and RCA

- **Corrective and Preventive Action (CPA):**
 - **Corrective Action:** Fixes existing problems (e.g., resolving customer complaints).
 - **Preventive Action:** Prevents potential problems (e.g., new protocols to avoid known errors).
 - **Steps:** Identify → Analyze → Plan → Implement → Monitor.
- **Root Cause Analysis (RCA):**
 - Identifies the underlying cause of a problem to prevent recurrence.
 - **Tools:** 5 Whys, Fishbone Diagram.
 - **Steps:** Define problem → Gather data → Analyze causes → Develop solutions → Monitor.

Example:

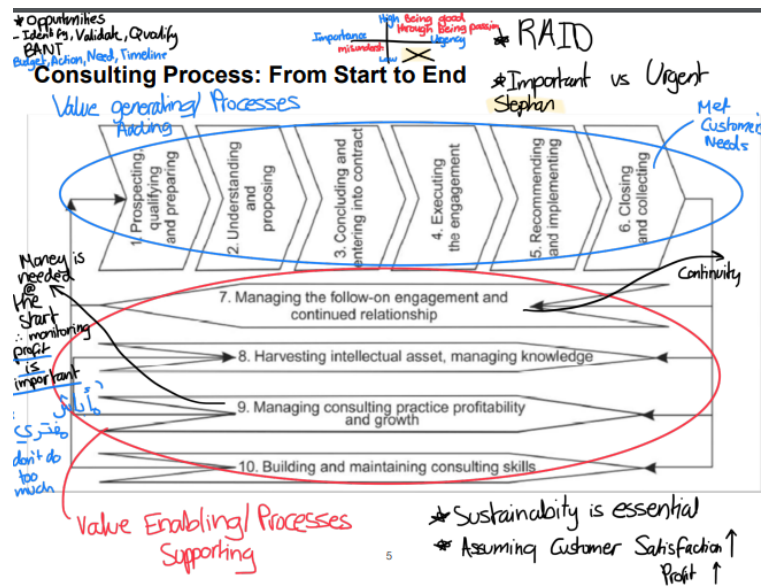
- **Problem:** Repeated delays in deliverables.
 - **RCA:** Lack of resource planning identified as the root cause.
 - **CPA:**
 - **Corrective:** Reschedule delayed projects.
 - **Preventive:** Implement better resource planning tools.

Tools and Artifacts:

- **Save time and enhance productivity.**
- **Examples:**
 - Templates for data collection.
 - Frameworks for problem analysis.
 - Diagnostic assessment tools.



Lecture Notes:



Start-to-End Process Steps:

Value Generating Processes:

These are the core consulting activities that directly contribute to delivering value to the client.

- 1. Prospecting and Preparing:**
 - Initial phase where opportunities are identified and validated.
 - Focus on understanding client needs, market conditions, and preparing for the engagement.
- 2. Understanding and Proposing:**
 - Involves deeply understanding the client's issues and formulating a proposal.
 - Customization to meet client needs and aligning with their expectations is critical.
- 3. Concluding and Entering into Contract:**
 - Formalizing the agreement through contract negotiation.
 - Ensures that both parties agree on the scope, deliverables, and terms.
- 4. Executing the Engagement:**
 - The main delivery phase where the consulting team implements the plan.
 - Success here depends on effectively managing resources and timelines.
- 5. Recommending and Implementing:**
 - Providing actionable recommendations and supporting their implementation.
 - Ensuring that the client can practically apply the solutions provided.
- 6. Closing and Collecting:**
 - Finalizing the engagement, ensuring all deliverables are met.
 - Involves collecting feedback, final payments, and any necessary post-project support.

Value Enabling (Supporting) Processes:

These processes support the consulting firm's internal capabilities and long-term sustainability.

- 7. Managing the Follow-on Engagement and Continued Relationship:**
 - Focus on client relationship management and follow-ups.
 - Ensuring a smooth transition and looking for future collaboration opportunities.
- 8. Harvesting Intellectual Asset, Managing Knowledge:**
 - Capturing lessons learned, methodologies, and insights.
 - Managing knowledge for reuse in future engagements.
- 9. Managing Consulting Practice Profitability and Growth:**
 - Financial management of the consulting practice.
 - Emphasizes profitability, cost control, and growth strategies.
- 10. Building and Maintaining Consulting Skills:**
 - Continuous professional development to enhance consulting capabilities.
 - Ensures that the team remains skilled and competitive.

Additional Notes and Annotations:

- **Continuity:** Emphasis on maintaining ongoing relationships and ensuring smooth transitions between engagements.
- **Sustainability:** Essential for long-term success, implying that practices should be designed to be viable over the long term.
- **Assuming Customer Satisfaction = Profit:** A note suggesting that achieving customer satisfaction is often equated with profitability, emphasizing the importance of meeting customer needs.
- **Met Customer's Needs:** A reminder of the goal to align consulting services with what the customer requires for success.

Key Annotations:

- **"Opportunities":** Emphasizes the need to identify and validate opportunities early.
- **"Name is important":** Importance of branding or reputation in consulting.
- **"Don't do too much":** Not overextending resources or trying to deliver beyond capacity.

RAID (Risks, Assumptions, Issues, Dependencies)

A project management tool used to track critical elements of a project to ensure its success.

- **Risks:** Potential problems that might occur and impact the project negatively.
- **Assumptions:** Factors considered true for planning but may need validation.
- **Issues:** Current problems that need resolution.
- **Dependencies:** Elements outside the project that influence its success.

Use: Helps identify, track, and mitigate challenges proactively.

Important vs. Urgent (Stephen Covey's Time Management Matrix)

Stephen Covey's framework for prioritizing tasks based on their importance and urgency.

1. **Quadrant 1: Important & Urgent (Crisis Zone)**
 - Tasks that require immediate action, like deadlines or emergencies.
Example: Fixing a critical system failure.
2. **Quadrant 2: Important & Not Urgent (Growth Zone)**
 - Long-term value tasks like planning, skill development, and relationship building.
Example: Strategic planning for a new project.
3. **Quadrant 3: Not Important & Urgent (Distraction Zone)**
 - Tasks that feel pressing but don't add significant value.
Example: Answering non-critical emails.
4. **Quadrant 4: Not Important & Not Urgent (Waste Zone)**
 - Unproductive activities.
Example: Excessive social media browsing.

Goal: Spend most time in Quadrant 2 to focus on proactive, impactful work.

BANT (Budget, Authority, Need, Timeline)

A sales qualification framework to assess potential customers' readiness to buy.

- **Budget:** Does the client have the financial resources?
- **Authority:** Is the person you're speaking with the decision-maker?
- **Need:** Does the client have a genuine need for your services?
- **Timeline:** When does the client want to implement the solution?

Sub-Processes

1. **Value-Enabling Sub-Processes:**
 - Support activities for relationship building that ensure consulting success.
2. **Value-Generating Sub-Processes:**
 - Activities tied directly to the execution of the engagement.

What a proposal (headings) would include:

1. **Background:** Provides context and reasons for the study.
2. **Objectives and Scope:** Defines project goals and boundaries, including exclusions to avoid ambiguity.
3. **Scope Details:** Covers both geographical and functional aspects.
4. **Issues to Address:** Lists the challenges the consulting team will tackle. (defects)
5. **Methodology:** Describes the approach and techniques to be used.
6. **Project Plan:** Outlines the timeline, milestones, and deliverables.
7. **Team Composition:** Details the roles of both consulting and client teams. Client teams are a requirement from the consultants to make the consulting exercise easier and faster to undertake

8. **Governance Structure:** Explains project management, monitoring, and issue resolution. How any disputes or any issues arising during the consultation phase will be addressed
9. **Responsibilities:** Clarifies roles of both consultant and client.
10. **Exclusions:** Specifies what is not included in the project. (Grey areas)
11. **Consulting Fee and Terms:** Provides financial details, payment terms, and conditions.

Additional Notes: Proposals can be in Word or PowerPoint, with fees detailed in a separate letter if needed.

Important Points from Quiz:

1. **Walkthroughs:** Regular meetings during engagements to review progress, identify issues, and ensure alignment.
 2. **Scope Changes:** Should not be informal or without additional cost considerations.
 3. **Value-Enabling Sub-Processes:** Aim to sustain the consulting firm by building capabilities like tools, frameworks, and relationship management.
 4. **Team Composition:** Includes members from both consulting and client organizations.
 5. **Progressive Elaboration:** Applies throughout the planning and execution phase of a project.
-

Topic 6: Consulting Engagements

Key Themes

- Planning, executing, and governing consulting engagements.
- Managing scope, customer acceptance, and engagement closure.
- Tools and techniques for consulting processes.

Keywords

- **Planning Engagement:** Search for team selection, kick-off meeting, objective-setting.
- **Executing Engagement:** Use walkthroughs, incremental sign-offs, and work-in-progress deliverables.
- **Engagement Governance:** Search for scope management, customer acceptance, and change processes.

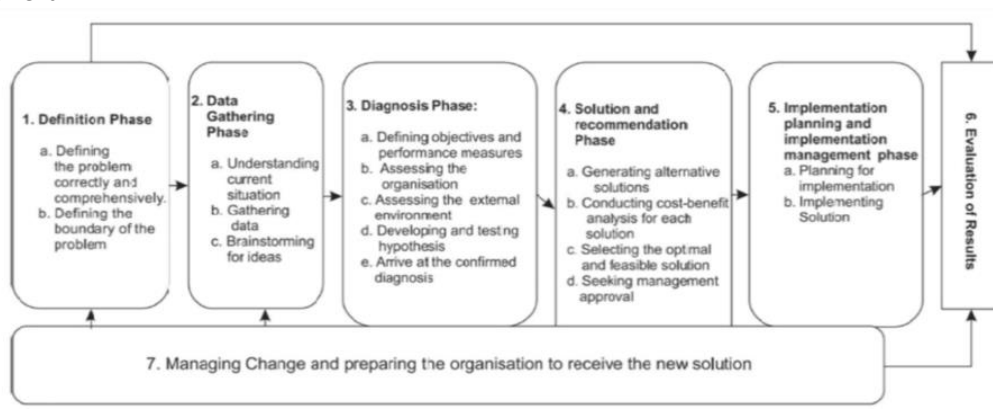
Introduction

- **Consultant's Role:** Helping clients move from the **current state** (where they are) to the **desired state** (where they want to be).
 - The engagement would be about guiding the client with any or all of the three elements: **The Current State, The Desired State, The Path from here to there**
 - **Path:** The method or steps to achieve the desired outcome.
- Consultants analyze problems from **various perspectives**, ensuring **holistic** (dealing with or treating the whole of something, not just a part) solutions. A successful consultant would choose the right set of tools, give recommendations using the simple, proven and available technology
- **Example:** Similar to a doctor diagnosing a patient with a fever, consultants use tools and proven methods to recommend solutions.

Planning the Engagement

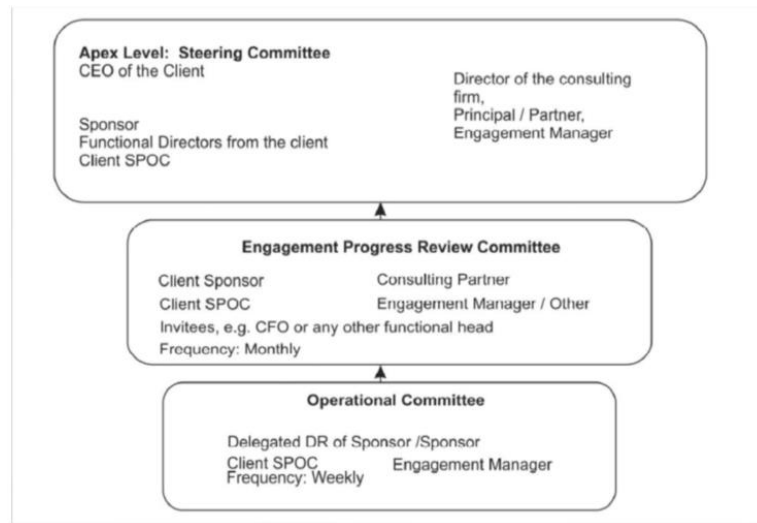
- Steps for effective planning:
 1. **Plan the Work:** Define tasks and responsibilities.
 2. **Team Selection:** Assemble the right mix of skills.
 3. **Brief The Team:** Provide clear instructions, context, and expectations to ensure all team members understand their roles and the project goals.
 4. **Give a starter pack to the team:** Provide essential resources and materials.
 5. **Gather suitable intellectual capital for use:** Use knowledge and tools for engagement.
 6. **Kick-off the engagement with the client:** holding an initial meeting to align on objectives, set expectations, and establish a collaborative working relationship for the project.
 7. **Kick-off Meeting:** Start the project with clear objectives and team alignment.
 8. **Team Building Exercise:** Foster collaboration and trust.
 9. **Objective-Setting Workshop:** Align on goals and expectations.

Executing the Engagement



- Focuses on delivering the plan, adapting to challenges, and ensuring alignment with client goals.
- Techniques:
 - Incremental progress reviews.
 - Work-in-progress deliverables.
 - Walkthroughs for feedback and alignment.

Engagement Governance



Apex Level Committees

Apex level committees are high-level groups responsible for overseeing and guiding the consulting engagement at different levels of responsibility.

1. Steering Committee

- **Purpose:** Provides strategic direction and ensures the project aligns with the organization's overall goals.
- **Example:** A group of senior executives overseeing a digital transformation project for a company. They decide on major issues like budget allocation or changes in project scope.
- **Key Role:** Approves high-level decisions, resolves escalated issues, and ensures alignment with business objectives.

2. Engagement Progress Review Committee

- **Purpose:** Monitors the progress of the consulting engagement and ensures milestones are being met.
- **Example:** A team of project sponsors and key stakeholders meets monthly to review the progress of a supply chain optimization project, ensuring deliverables are on track.
- **Key Role:** Tracks project status, addresses mid-level challenges, and facilitates communication between the operational team and the steering committee.

3. Operational Committee

- **Purpose:** Handles the day-to-day execution of the project, ensuring tasks are completed as planned.
- **Example:** A team of managers and consultants working on implementing a new CRM system for a company. They handle scheduling, task assignments, and immediate problem-solving.
- **Key Role:** Focuses on operational details, reports progress to the review committee, and manages team-level issues.

Managing the Scope and Customer Acceptance

Guidelines for Handling Potential Scope Creep

- **Scope creep:** The gradual expansion of a project's scope beyond its original boundaries, often leading to delays or budget overruns.

Avoid an Upsetting "No"

- **Explanation:** Instead of outright rejecting additional requests, discuss alternative ways to address them.
- **Example:** If a client requests a feature outside the initial agreement, explain the additional resources or time required rather than refusing outright.

Think Jointly with the Requestor in Terms of Resource and Time Needs

- **Explanation:** Collaborate with the client to evaluate the feasibility of their request, considering available resources and timelines.
- **Example:** Work with the client to prioritize the requested feature and decide whether it can fit into the current phase or needs to be deferred.

Always Course the Changes Through a Formal Process

- **Explanation:** Ensure that any modifications to the scope are documented, approved, and integrated into the project plan.
- **Example:** Use a formal change request process to assess impacts on budget, timelines, and deliverables before proceeding.

Avoid “Big-Bang” Signoffs

- **Explanation:** Do not wait until the end of the project to seek client approval for all deliverables at once.
-

Get Incremental Signoffs

- **Explanation:** Obtain client approval at key milestones to confirm alignment and satisfaction with progress.
 - **Example:** After completing a prototype, get the client’s feedback and signoff before moving to the next stage.
-

Present “Work-in-Progress” Deliverables

- **Explanation:** Share partially completed work with the client to gather feedback and make adjustments early.
 - **Example:** Show a draft version of a report or initial designs for input before finalizing.
-

Conduct Walkthrough Reviews Whenever Appropriate

- **Explanation:** Regularly review the project’s progress with the client to identify and resolve issues collaboratively.
- **Example:** Schedule weekly meetings to discuss updates, address concerns, and ensure the project remains on track.

Engineering Terms Explained

- **Scope Creep:** Gradual expansion of a project’s scope beyond the initial agreement.
 - **Incremental Sign-offs:** Regular approvals for parts of the project, ensuring client satisfaction at each stage.
-

Topic 7: Introduction to Business Analysis (BA)

Key Themes:

- Introduction to Business Analysis (BA).
- Origins and development of BA.
- Business analyst role development and range of analysis activities.
- BA principles and maturity model.
- The BA Service Framework (BASF).
- Variants of the business analyst role.
- Professionalism in BA.
- Future of BA.

Key Words for Quick Reference:

- Origins of BA: IT evolution, outsourcing gaps.
- BA Principles: Root causes, holistic view, options.
- BA Maturity: CMMI, professional standards.
- Roles: Enterprise BA, Digital BA.
- Service Framework: BASF, BA techniques.

Introduction:

Business Analysis ensures that:

- Business changes align with organizational needs.
- A holistic (**dealing with or treating the whole of something, not just a part**) approach is taken, considering all relevant aspects.

BA is an important discipline that has evolved over decades, offering services to enhance business processes, align IT with business goals, and implement innovative strategies.

Origins of Business Analysis:

- IT initially focused on building systems to improve business operations and management decision-making.
- As business priorities shifted, IT evolved to support:
 - Development of new services and products.
 - Flexible communication mechanisms that enabled innovative business models.
- Technology created opportunities for organizations to focus on their core capabilities without being distracted by peripheral (external) areas of business requiring specialist capability. Outsourcing IT services revealed gaps in supplier management and requirement definitions, necessitating BA roles.
- Despite these advances, dissatisfaction grew due to the inability of technology and change disciplines to fully support businesses, highlighting the need for specialized BA functions.

Development of BA:

The impact of outsourcing (hiring a party outside a company to perform services)

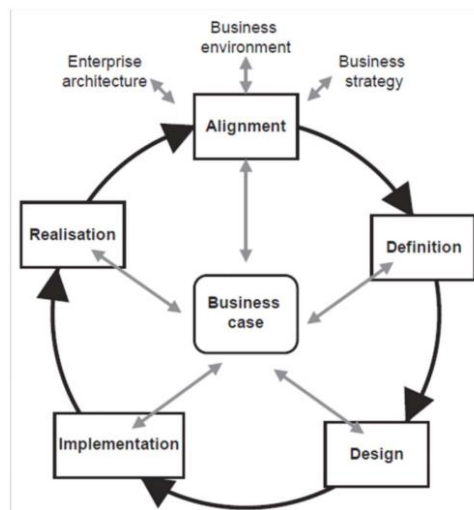
Many organizations outsourced IT services to reduce costs, leading to challenges such as:

- **Supplier management issues:** Often handled by procurement function and Project Management Offices (PMOs).
 - **Procurement team:** responsible for buying and acquiring the goods the company sells. It interacts with suppliers and manufacturers to procure (obtain) items and services that provide the best value for money.
- **Requirement definition issues:** Handled by business analysts to ensure clarity and alignment of requirements.

Competitive advantage using IT

1. **IT Systems and Competitive Advantage:**
 - **For IT to provide a competitive edge, three factors are crucial:**
 1. Business needs must drive IT development.
 2. IT implementations must include necessary business changes.
 3. IT system requirements must be clearly defined and understood.
2. **Business Change Lifecycle:**
 - **Effective BA involves analyzing and managing the entire lifecycle of business changes to ensure value realization.**

The business change lifecycle



1. **Definition:**
 - **Focus:** Clarify the problem, define business needs, and establish objectives for the change.
 - **Connection to Business Case:** Inputs define scope and objectives, while outputs refine and justify the business case.
2. **Design:**
 - **Focus:** Develop a detailed blueprint for the solution, including processes, systems, and roles.
 - **Connection to Business Case:** Inputs ensure the design aligns with objectives, while outputs validate feasibility and benefits.
3. **Implementation:**
 - **Focus:** Execute the designed changes, including system deployment and process improvements.
 - **Connection to Business Case:** Inputs guide implementation priorities, and outputs track progress and alignment with expected outcomes.
4. **Realization:**
 - **Focus:** Measure and validate that the implemented changes deliver the expected business benefits.
 - **Connection to Business Case:** Inputs measure outcomes against the business case, while outputs update the case with realized benefits.
5. **Alignment:**
 - **Focus:** Ensure ongoing alignment of changes with the enterprise architecture (EA), business environment, and overall strategy.
 - **Connection to Business Case:** Inputs include strategic shifts and external factors; outputs ensure the case remains relevant and adaptive to the organization's strategy and environment.

This cyclical model shows that the business case is both a guiding framework and a dynamic document updated throughout the lifecycle.

Business Analyst Role Development:

Support for Business Managers:

- Business managers increasingly rely on BAs to analyze necessary changes and use investment funds prudently (in a way that is careful and avoids risks) in challenging economic conditions.
- The role involves not just executing projects but contributing to strategic decision-making.

Challenges in Defining the Role:

- Misconceptions and lack of clarity persist about the BA role.
- Common questions include:
 - What do BAs do?
 - What skills do they need?
 - How do they add value?

The Business Analysis Service Framework (BASF) proposes an outline service catalogue that encompasses (include as part of something broader):

1. **Services Provided by Business Analysts:**
 - **Definition:** Specifies the range of services BAs offer, such as process improvement, requirements gathering, and change management.
 - **Value Proposition:** Clearly outlines the benefits and outcomes of each service to demonstrate its contribution to business goals.
2. **Activities Conducted by Business Analysts:**
 - **Purpose:** Details the tasks and actions performed to deliver each service effectively.
 - **Examples:** Includes stakeholder interviews, workshop facilitation, business process modeling, and gap analysis.
3. **Key Techniques Used in Service Delivery:**
 - **Focus:** Describes the methodologies and tools BAs employ to execute their tasks.
 - **Examples:**

- Unified Modeling Language (UML) for visual modeling.
 - SWOT analysis for evaluating strengths, weaknesses, opportunities, and threats.
 - Agile techniques like user story mapping.
- **Emerging Techniques:** Highlights that new tools and methods are continually developed to adapt to evolving business needs.

The Range of Analysis Activities:

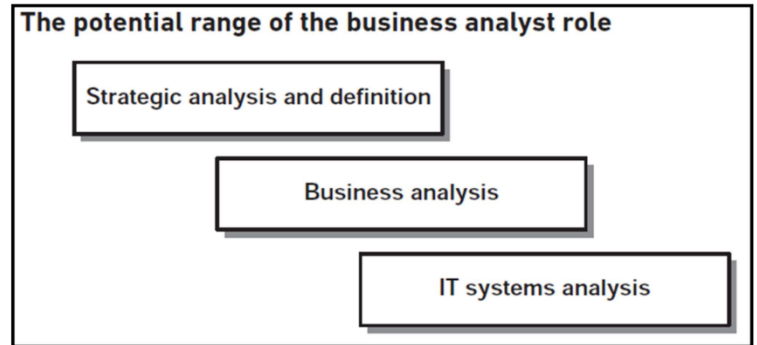
- Business analysis lies between two ends of the spectrum:

1. Strategic Analysis:

- Focuses on high-level business strategy, often involving senior management and consultants.
 - BAs need to understand strategic outcomes to align their work.
 - Often have to recommend, design and deploy the tactics that are needed to execute the business objectives and strategy (typically concern the process and IT systems solutions)
 - and should be able to work within the strategic context for their organizations.
 - The business applications and data architectures are of particular relevance to business analysts as they need to ensure that these architectural domains support and align with any business change proposals.

2. IT Systems Analysis: (System Analysts: used less often these days)

- Systems analysts (**not BA**) are responsible for analyzing and specifying the IT system requirements in sufficient detail to provide a basis for the evaluation of software products or the development of an IT system.
 - The essential difference between business analysis and IT systems analysis is that a business analyst has a business outcome focus and a systems analyst has a software focus.
 - The distinction is not always clear and the advent (beginning) of Agile approaches has often resulted in the restructuring of IT and change roles within organizations.



Between these, business analysts work to:

- Investigate **localized issues or broad business challenges** that require investigation into several issues or ideas for increased efficiency or effectiveness.
- enhance or replace an **existing software product** to meet new business requirements.
- Analyze software products or entire workflows for improvement using a **service view and value realization approach**.
- Business analysis assignments start with understanding the current business situation.
- Initial problem definitions and proposed solutions **often do not capture all underlying issues**.
- Most cases involve a broad range of problems requiring extensive investigation.
- Clarifying the actual problem is often the critical first step to avoid examining irrelevant areas.
- A **holistic approach** is essential, considering:
 - Business processes,
 - Software products,
 - Job roles, and
 - Employee skills.
- Key services delivered in such assignments include:
 - **Situation investigation and problem definition**,
 - **Business process improvement**, and
 - **Requirements definition** to identify and implement necessary changes.

BA Principles:

1. Root Causes, Not Symptoms

- **Explanation:**
 - Focus on identifying and addressing the underlying causes of problems instead of just resolving visible symptoms.
 - Take a holistic view to ensure all relevant aspects of the issue are considered.
- **Example:**
 - If customer complaints about delays are rising, investigate root causes like inefficient supply chain processes or inadequate staffing rather than just improving customer service scripts.

2. Business Improvement, Not IT System Change

- **Explanation:**
 - IT systems should support business opportunities and problem-solving, not be the sole focus of change.
 - Analyze how business improvements can drive innovation and enhance customer experiences.
- **Example:**
 - Instead of upgrading an IT system for better reporting, analyze how streamlined processes or automation can improve the entire decision-making workflow.

3. Options, Not Solutions

- **Explanation:**
 - Challenge pre-determined solutions.
 - Identify and evaluate multiple options to meet business needs effectively.
- **Example:**
 - If a client proposes implementing a specific CRM software, explore other tools or approaches to see if they better align with business goals or budgets.

4. Feasible, Contributing Requirements, Not All Requests

- **Explanation:**
 - Prioritize requirements based on feasibility and their contribution to business objectives, considering constraints like time and cost.
 - Avoid implementing requirements that don't add value.
- **Example:**
 - Decline a request for a low-priority feature in a new application if it delays delivery or exceeds budget, focusing instead on essential functionality.

5. The Entire Business Change Lifecycle, Not Just Requirements Definition

- **Explanation:**
 - Support the complete lifecycle of change, from analyzing the situation to post-implementation review and benefit realization.
- **Example:**
 - After deploying a new order management system, monitor performance, train employees, and conduct post-launch reviews to ensure business goals are met.

6. Negotiation, Not Avoidance

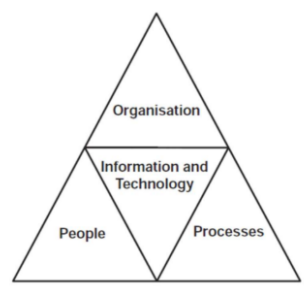
- **Explanation:**
 - Address and reconcile conflicting stakeholder views and requirements through effective negotiation.
- **Example:**
 - If marketing wants more personalization features in a product while IT focuses on security, negotiate a solution that balances both priorities.

POPIT:

1. **People:**
 - Ensure individuals have the skills, training, and support needed for changes while addressing cultural attitudes and resistance.
2. **Organization:**
 - Align the company's structure, strategy, and leadership with business goals and readiness for change.
3. **Processes:**
 - Identify inefficiencies in workflows and redesign processes to improve efficiency and value delivery.
4. **Information:**
 - Ensure data is accurate, accessible, and flows seamlessly across teams and systems to support decisions.
5. **Technology:**
 - Evaluate IT systems for alignment with business needs and scalability for future growth.

Holistic Approach:

By considering all components of the POPIT model together, business analysts ensure that changes are effective, sustainable, and address all areas impacted by the initiative. Ignoring any one element can lead to incomplete solutions and potential failures.



Holistic approach

Agile Approach:

Better ways of developing software by doing it and helping other do it. Through this work we have come to value:

- **Individuals and interactions** over processes and tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

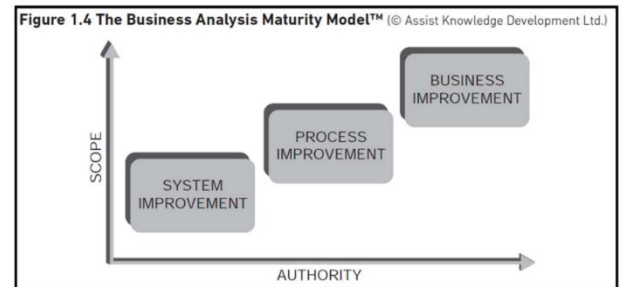
While there is value to items on the right, we value items on the left more

The BA Maturity Model:

Business Analysis Maturity Model (Scope vs. Authority)

This figure illustrates three levels of business analysis maturity based on scope and authority within an organization. The levels progress as follows:

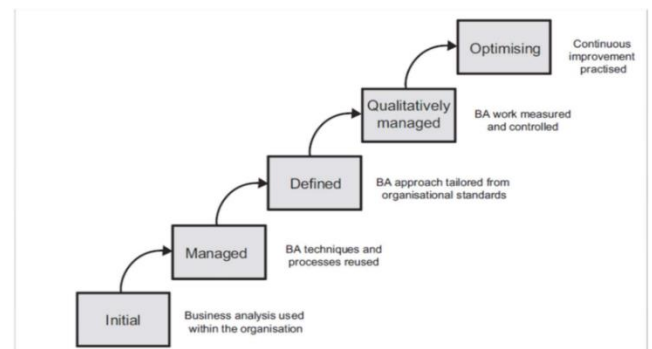
1. **System Improvement:**
 - Focus is limited to individual systems or technology components.
 - Authority is restricted to specific systems or projects.
 - Example: An IT team updates a customer relationship management (CRM) system to improve its user interface.
2. **Process Improvement:**
 - Broader scope, addressing end-to-end processes across departments.
 - Authority includes driving changes to operational workflows.
 - Example: Redesigning the sales order process to integrate CRM, inventory, and billing systems for greater efficiency.
3. **Business Improvement:**
 - Organization-wide scope with authority to influence strategic decisions.
 - Emphasis on aligning business strategy with operational changes.
 - Example: Implementing an enterprise-wide digital transformation strategy to enhance customer experience and market competitiveness.



Business Analysis Capability Maturity Model (Phases)

This figure represents the progression of an organization's business analysis capability in five maturity levels:

1. **Initial:**
 - Business analysis practices are ad hoc (case-by-case/temporary) and inconsistent.
 - Example: Individual teams occasionally use business analysis for projects without standard processes.
2. **Managed:**
 - Basic business analysis techniques and processes are used consistently.
 - Example: A defined template for gathering project requirements is adopted across projects.
3. **Defined:**
 - Business analysis practices are tailored to organizational standards.
 - Example: A retail company creates a custom methodology for analyzing customer feedback during project planning.
4. **Quantitatively (Not Qualitatively as in the slides, changed by the doctor) Managed:**
 - Business analysis outcomes are quantitatively measured and controlled to ensure consistent quality.
 - Example: A healthcare provider tracks and evaluates the effectiveness of business analysis activities by measuring how closely project requirements align with the final deliverables, using specific metrics such as requirement traceability, defect rates, or stakeholder satisfaction scores.
5. **Optimizing:**
 - Continuous improvement of business analysis practices is prioritized.
 - Example: A technology company reviews BA processes quarterly to identify areas for improvement and adopts emerging tools like AI for data analysis.



- Derived from **Capability Maturity Model Integration (CMMI)**.
- Defines the evolution of BA practices within organizations.

The Business Analysis Service

- A defined service portfolio with clear value propositions.
- The realization of value from business analysis imposes responsibilities on customers as well as the business analysts.



- Includes:
 - Core Services:
 - Situation investigation and problem analysis.
 - Requirements definition.
 - Feasibility assessment and business case development (evaluating the practicality and viability of proposed business changes and creating a structured justification, outlining costs, benefits, risks, and alignment with organizational goals)
 - Business process improvement.
 - Business acceptance testing
 - Business change deployment
 - Extended Services:
 - Strategy analysis and execution
 - The business analysts work closely with senior management to help define the most effective business system to implement elements of the business strategy.
 - IT requirements specification
 - The business analysts use modelling techniques, such as those from the Unified Modeling Language (UML), to specify functional and non-functional requirements.

Variants of Business Analyst Roles

- **Business/ Enterprise BA:** Aligns BA with organizational goals.
- **Technical BA/ Business Systems analyst:** Focuses on IT system specifications.
- **Digital BA:** Addresses digital transformation needs.
- **Project BA:** Embedded in specific projects.
- **Proxy Product Owner:** Represents stakeholder needs in Agile setups.

Professionalism in Business Analysis

- Professional body like BCS and IIBA.
- Certificates:
 - BCS Diplomas.
 - IIBA (e.g., CBAP®, CCBA®, ECBA™).
- Specialist Qualifications:
 - (IIBA-AAC, IIBA CBDA)
- Techniques and documentation standards and continuous professional development maintain professionalism.

(IIBA Certified Business Analysis Professional (CBAP®), Certificate of Capability in Business Analysis (CCBA®) and Entry Certificate in Business Analysis (ECBATM), and specialist qualifications including the Agile Analysis Certification (IIBA-AAC) and Certification in Business Data Analytics (IIBA CBDA))

Future of BA:

- Growing demand for roles integrating business and IT expertise.
 - Emphasis on soft skills, negotiation, and managing conflicting stakeholder views.
 - **Stakeholder Engagement:** Collaboration, emotional intelligence, cultural awareness, and teamwork to work effectively with people.
 - **Analytical Thinking:** Data analysis, evaluation, prioritization, and interpretation skills.
 - **Innovative Thinking:** Critical thinking, root cause analysis, creativity, problem-solving, and adaptive thinking.
 - **Technical Literacy:** Understanding and applying technological advancements like AI, virtual reality, and robotics.
 - **Value Co-Creation:** Achieving beneficial outcomes using service, design, lean, and systems thinking approaches.
-

Key Concepts from Quiz:

- **Learning Organization** as described by Peter Senge in his book “The Fifth Discipline” is Defined by shared vision and lifelong learning
 - **Business Analysis Services Framework** **DONOT** cite the monetary (related to money) value of each service
 - **Business Analysis Maturity Model** is based on Capability Maturity Model Integration.
 - **BA** focus on soft and cognitive skills more than technical
-

Topic 8: Fundamentals of Domain-Driven Design (DDD)

Key Themes

- Ingredients of Effective Modeling
- Knowledge Crunching
- Continuous Learning
- Knowledge-Rich Design
- Deep Models
- Ubiquitous Language
- Model-Driven Design
- Why Models Matter to Users

Keywords for Quick Reference:

- **Effective Modeling:** Involves binding the model to implementation, cultivating a domain-specific language, developing a knowledge-rich model, and refining through brainstorming
 - **Domain Model:** For design-user alignment.
 - **Ubiquitous Language:** To address communication issues.
 - **Deep Models:** For abstraction-focused questions.
 - **Knowledge Crunching:** Relevant for learning and collaboration topics.
 - **Model-Driven Design:** Unified modeling techniques.
 - **User-Centric Models:** Highlight the importance of aligning the system's design with user expectations to avoid confusion and errors.
-

Ingredients of Effective Modeling

1. **Binding the Model and the Implementation:**
 - Models must directly shape the software’s code, ensuring alignment between domain understanding and implementation.
 2. **Cultivating a Language Based on the Model:**
 - Shared language minimizes ambiguity (**unclear**) between team members and stakeholders.
 - **Example:** In a payment processing system, defining terms like "transaction" explicitly allows the team to communicate clearly about financial processes.
 3. **Developing a Knowledge-Rich Model:**
 - Models include not just entities (e.g., products, orders) but also the business processes and rules governing them.
 4. **Distilling the Model:**
 - Simplification is key. Overly detailed models can lead to inefficiency.
 5. **Brainstorming and Experimenting:**
 - Iteration uncovers better models. Brainstorming with domain experts may lead to new abstractions.
-

1. Knowledge Crunching

- **Concept:**
 - **Iterative Refinement for Simplicity:** Effective domain modelers continuously explore and test organizing ideas, aiming to uncover simple, abstract concepts that make sense of the complexity within the domain. This process involves distilling vast details into a focused and meaningful representation of the most relevant knowledge.
 - **Collaborative Knowledge Building:** A team effort, led by developers working closely with domain experts, ensures that the refined model captures the essential business rules and processes, blending technical precision with domain-specific insights.
 - **Mutual Growth Through Collaboration:** As team members collaboratively refine the domain model, developers deepen their understanding of critical business principles rather than to produce functions mechanically., while domain experts clarify and distill their knowledge, fostering shared insights and conceptual rigor(precise/detailed). **(The precise approach to defining, organizing, and understanding concepts to ensure clarity, consistency, and logical accuracy.)**
-

2. Continuous Learning

1. **The Challenge of Knowledge Loss:**

- Knowledge is inherently fragmented(separated), scattered among people and documents. It risks being lost during team transitions.
 - **Hidden Complexity and False Assumptions:** Seemingly simple domains often mask unknown complexities, leading to false assumptions that affect the project's direction.
 - **Knowledge Loss and Fragmentation:** Knowledge leakage occurs as team members leave or reorganize, and outsourcing can result in delivered code without the critical knowledge necessary for long-term use.
2. **Building a Stable Core Team:**
 - Retaining domain knowledge within a stable core of team members ensures continuity.
 - The accumulated knowledge in the minds of this core team makes them more effective knowledge crunchers.
 3. **PCB Design Tool Illustration:**
 - Highlighting domain-specific understanding, a tool's design improves as developers grasp the intricacies (quality of being complex) of printed circuit board layouts.

3. Knowledge-Rich Design

1. **Central Role of Business Rules:** Business activities and rules are as critical to the domain as its entities, and effective knowledge crunching creates models that capture these insights, requiring developers to refactor implementations accordingly.
2. **Navigating Complexity and Inconsistencies:** Business rules can be inconsistent, and domain experts often handle these complexities unconsciously, relying on intuition and common sense to reconcile (resolve) contradictions.
3. **Collaboration for Clarity:** Close collaboration between domain experts and developers during knowledge crunching helps clarify, refine, and address these rules, ensuring the model aligns with domain needs.
4. **Clarifying Complexity through Collaboration:**
 - Close developer-expert collaboration identifies and resolves contradictions in business rules.
 - **Example:** A supply chain model might clarify rules about when "backorders" are acceptable versus when "expedited shipping" is required.

Example from lecture:

Let's start with a very simple domain model that could be the basis of an application for booking cargos onto a voyage of a ship.

```

classDiagram
    Voyage "1" -- "*" Cargo
  
```

We can state that the booking application's responsibility is to associate each Cargo with a Voyage, recording and tracking that relationship. So far so good. Somewhere in the application code there could be a method like this:

```

public int makeBooking(Cargo cargo, Voyage voyage) {
    int confirmation = orderConfirmationSequence.next();
    voyage.addCargo(cargo, confirmation);
    return confirmation;
}
  
```

The requirements document contains this line:

Allow 10% overbooking.

The class diagram and code now look like this:

```

classDiagram
    class Voyage {
        capacity
        bookedCargoSize
    }
    class Cargo {
        size
    }
    Voyage "1" -- "*" Cargo
    Voyage --> Voyage : (sum(cargo.size()) + cargo.size()) > maxBooking
  
```

```

public int makeBooking(Cargo cargo, Voyage voyage) {
    double maxBooking = voyage.capacity() * 1.1;
    if ((voyage.bookedCargoSize() + cargo.size()) > maxBooking)
        return -1;
    int confirmation = orderConfirmationSequence.next();
    voyage.addCargo(cargo, confirmation);
    return confirmation;
}
  
```

The code is now:

```

public int makeBooking(Cargo cargo, Voyage voyage) {
    if (!overbookingPolicy.isAllowed(cargo, voyage)) return -1;
    int confirmation = orderConfirmationSequence.next();
    voyage.addCargo(cargo, confirmation);
    return confirmation;
}
  
```

The new **Overbooking Policy** class contains this method:

```

public boolean isAllowed(Cargo cargo, Voyage voyage) {
    return (cargo.size() + voyage.bookedCargoSize()) <=
        (voyage.capacity() * 1.1);
}
  
```

4. Deep Models

1. **Evolving Understanding of the Domain:** Initial models that seemed important at the beginning often include superficial elements that are later discarded or reframed (we shift their perspective) as deeper insights emerge during domain exploration.
2. **Subtle abstractions** emerge that would not have occurred to us at the outset but that pierce to the heart of the matter. **(The most important or central issue, without unnecessary details or distractions.)**
3. **From Surface to Core Abstractions:** In a container shipping system, the initial model focused on physical aspects like cargo and itineraries (a plan of a journey/ route) but evolved to emphasize responsibility transfers between parties.
4. **Shift in Perspective:** Designers realized that the core of the domain lay in the legal and practical transfer of responsibility, not in the physical handling of cargo (which was largely carried out by subcontractors or by operational people in the company.), fundamentally changing the model.
(This deeper view of the shipping business did not lead to the removal of the Itinerary object, but the model changed profoundly)
5. **Enhanced Model Support:** The refined model aligned operations with responsibility transfers **(transferring responsibility for cargo from entity to entity rather than moving containers from place to place)**, replacing awkward implementations with a cohesive structure reflecting these relationships.

5. Ubiquitous (Being Everywhere) Language

Ubiquitous Language

- **Definition:** A shared language between domain experts and developers to ensure everyone understands the same terms in the same way.
- **Problem:**
 - Domain experts know the business jargon (terminology) but not the technical terms.
 - Developers know technical jargon but lack deep domain understanding.
 - Without a common language, communication is inefficient and prone to misunderstanding.
 - A few members of the team manage to become bilingual (understand both jargons), but they become bottlenecks of information flow, and their translations are inexact.
 - Without a common language, developers and domain experts must constantly translate for each other, causing confusion and miscommunication. This leads to muddled (confusing) model concepts and frequent, potentially harmful refactoring of the code.
- **Solution:**
 - Develop a common language based on the domain model.
 - **More robust than the lowest common denominator:** a language that goes beyond the simplest or most basic terms that everyone understands. Instead, it aims to develop a comprehensive and precise language that fully captures the complexity of the domain, ensuring clear and effective communication without oversimplification.
 - Use the model as the foundation for the language.
 - Ensure the team consistently uses this language in all communication and code.
 - Experiment with alternative expressions to refine the language and model, then refactor the code accordingly.
 - Treat changes in the ubiquitous language as changes to the model, addressing any awkward or unclear terms to maintain clarity and consistency.
- **Benefits:**
 - Reduces translation errors between domain experts and developers.
 - Makes the code easier to understand and maintain.

Ubiquitous Language

- **Implementation:**
 - Incorporate the language into class names, methods, and documentation.
 - Refactor code to align with the evolving language.
 - Use diagrams and writing to reinforce the common language.
- **Feedback Loop:**
 - As the language is used, identify and fix awkward or unclear terms.
 - This process refines both the language and the underlying domain model.

Example Scenarios

- **Scenario 1:**
 - **Situation:** Changing the customs clearance point in a cargo routing system.
 - **Developer's Response:** Describes the technical process, including database operations.
 - **User's Concern:** Clarifies practical implications, like unnecessary extra work.
- **Scenario 2:**
 - **Situation:** Adjusting the route specification attributes.
 - **Developer's Plan:** Automatically regenerate the itinerary unless it still satisfies the new specification.
 - **User's Feedback:** Avoid unnecessary rerouting unless a change is necessary.

6. Model-Driven Design

- **Traditional Approach:**
 - **Analysis Model:** Created to understand the business domain, focusing solely on organizing concepts without considering software implementation.
 - **Design Model:** Created later, often with little connection to the analysis model, leading to inefficiency.
- **Model-Driven Design (MDD):**
 - Discards the dichotomy (separation) between analysis and design models.
 - **Single Model:** Serves both understanding the domain and driving the software design.
 - **Terminology and Responsibilities:** Drawn directly from the model, ensuring consistency between design and implementation.
 - **Ripple Effect:** Changes in the code reflect changes in the model and vice versa.

Why Models Matter to Users

- **Mismatch Consequences:**
 - An interface that does not align with the underlying model can confuse users and introduce bugs.
- **User Model vs. Design Model:**

- **User Model:** How the user perceives and interacts with the system.
 - **Design Model:** How the system is actually structured and functions.
 - **Consistency:** A model-based design reflecting the users' concerns provides a more predictable and understandable system.
 - **Revealing the Model:** Exposing the underlying design helps users utilize the software more effectively and predict its behavior.
-

Quiz Insights

1. **The "miss the wood for the trees" syndrome** refers to focusing too much on small details while failing to see the bigger picture or overall context.
 2. **DDD** handles both agile and waterfall methodologies
 3. **DDD** is **NOT** employed universally, it is employed according to the complexity of the domain.
 4. **PRIMARY purpose of using a Ubiquitous Language:** promote continuous refinement of the underlying domain model.
-

Midterm Info:

- Defining exclusions in a consulting proposal limits the consultant's liability for tasks outside the project's scope.
 - Setting realistic expectations with the client about the timeline and progress helps manage their goals effectively.
 - Employers are increasingly valuing candidates' soft and cognitive skills alongside technical competencies.
 - Starting with an analysis of the existing processes ensures a more accurate assessment of the feasibility of automation goals.
 - Constant refinement of the domain model helps developers align software solutions closely with business objectives.
 - Non-functional requirements like performance, security, and usability are critical for ensuring software meets business needs.
 - Prudent(Carefulness) use of funds is crucial to avoid jeopardizing an organization's ability to invest in essential areas during economic uncertainty.
 - Intellectual capital in consulting involves reusable patterns and approaches adaptable to different contexts.
 - In the context of business analysis principles, prioritizing achievable requirements within constraints ensures alignment with business objectives.
 - The V-shaped skill model categorizes skills based on technical depth (vertical) and broad knowledge (horizontal).
 - Knowledge transfer involves various methods like mentoring, workshops, documentation, and collaborative tools.
 - False assumptions about less complex domains can result in inaccurate domain models that fail to capture business requirements.
 - BANT stands for Budget, Authority, Need, and Timing, used in qualifying business opportunities.
-

Topic 9: Introduction to Domain-Driven Design (DDD)

Key Themes:

- Context Mapping
- Application Architecture
- Principles for Teams Starting with DDD
- Applying DDD Principles, Practices, and Patterns

Keywords:

- **Context Mapping:** Use for relationships, bounded contexts, and ACLs.
- **Layered Architecture:** Search when discussing software organization or dependency inversion.
- **Ubiquitous Language:** Reference for shared terms and communication.
- **Anti-Corruption Layer:** Relevant for translating and isolating contexts.

1. Context Mapping

Definition and Importance

- **Context Mapping:** A technique to define and visualize relationships between bounded contexts (specific parts of a domain that have their own models) and their interactions.
- Vital for understanding how different parts of a system communicate and ensuring boundaries are respected.

Model Relationships

- Defines relationships between **bounded contexts** (self-contained areas of a domain with clear boundaries).
- Encourages **explicit contracts** to clarify communication and dependencies.

Organizational Relationship Patterns

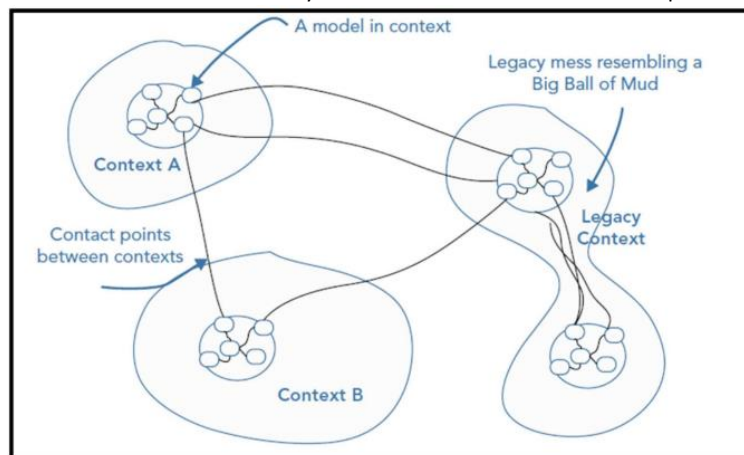
- Explains relationships between teams responsible for different contexts:
 - **Customer-Supplier:** Clear upstream and downstream roles.
 - **Conformist:** Adopts upstream models to avoid miscommunication.
 - **Anti-Corruption Layer:** Insulates a model from external influences by translating between systems.

Communication

- Effective communication requires maintaining context maps using clear visual tools (e.g., diagrams).

Key Visuals

- **Reality Map:** A visual representation of the actual state of a system's contexts and their relationships.



Context:

- **A bounded context**, a specific, self-contained area of a domain (area of work or problem like managing orders in an online store) with its own model, rules, and language.
- **Purpose:** It isolates different parts of a system to prevent confusion and maintain clarity.
- **Example:** In an e-commerce system, the **Order Management** and **Inventory Management** modules could be separate contexts.

A Model in a Context

- **A structured representation** of the domain logic, including entities, processes, and rules relevant to that context.
- **Purpose:** Provides a shared understanding for developers and domain experts.
- **Example:** In the **Order Management** context:
 - Entities: Order, Customer, Payment.
 - Processes: Order validation, payment processing.
 - Rules: Orders can only be processed if inventory is sufficient.

Legacy Context

- **Definition:** A **legacy context** is a system or component built with older, possibly outdated technologies or practices, often lacking modularity and modern standards.
- **Characteristics:**
 - Often monolithic and hard to maintain.
 - Poorly documented, making integration challenging.
- **Example:** A mainframe-based system managing customer records that is tightly coupled with older applications.

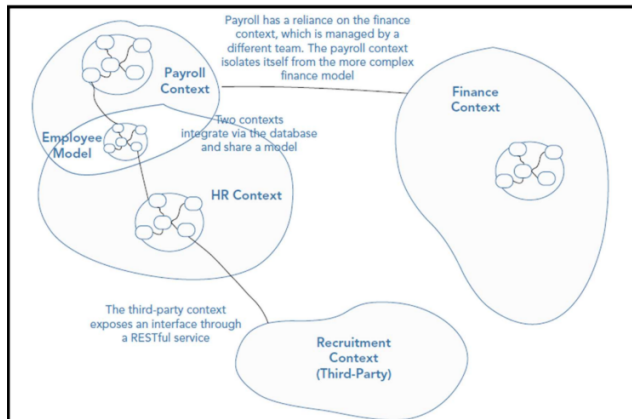
Big Ball of Mud

- A **Big Ball of Mud** describes a disorganized, monolithic codebase with:
 - Poor separation of concerns.
 - Undefined boundaries between components.
 - A lack of clear design principles.
- **In a Reality Map:**
 - It represents an **undesirable state** where a system lacks modularity and introduces high cognitive load for developers.
- **Example in Practice:**
 - A single system handling orders, payments, and inventory with tightly coupled code, making changes to one area risky and unpredictable.

How Contexts Are Represented:

- Well-defined contexts appear as distinct, modular areas with clear communication protocols.
- Legacy contexts or **Big Ball of Mud** systems are shown as tangled areas with unclear boundaries, representing integration difficulties and risks.

Technical Reality:



Organization Reality:

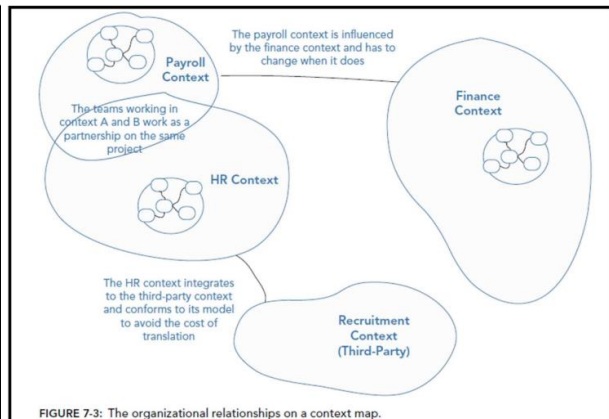


FIGURE 7-3: The organizational relationships on a context map.

The **technical reality** focuses on how systems, software, and technologies interact, while the **organizational reality** deals with how teams, roles, and workflows are structured and collaborate.

- **Anti-Corruption Layers (ACLs):** Protect domain logic from upstream influence.

Example:

Scenario: A company has a modern inventory system but still needs to integrate with an old legacy system for processing orders.

Solution:

- An ACL is created to act as a translator between the two systems.
- The modern system sends requests in its own format (e.g., JSON with clear field names), but the ACL converts these into the format the legacy system understands (e.g., XML with obscure field names).
- Similarly, responses from the legacy system are converted back by the ACL into the modern system's format.

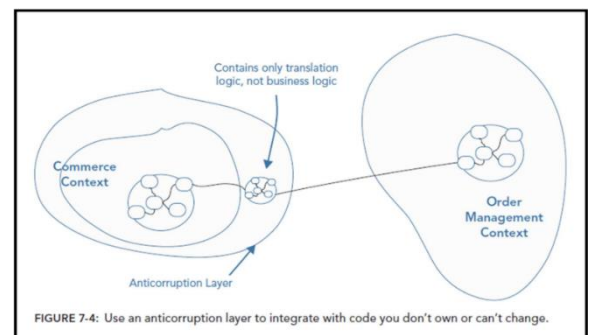
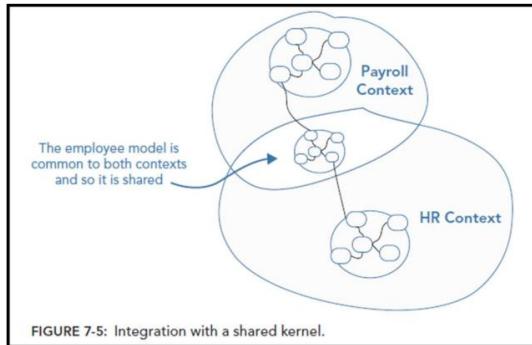


FIGURE 7-4: Use an anticorruption layer to integrate with code you don't own or can't change.

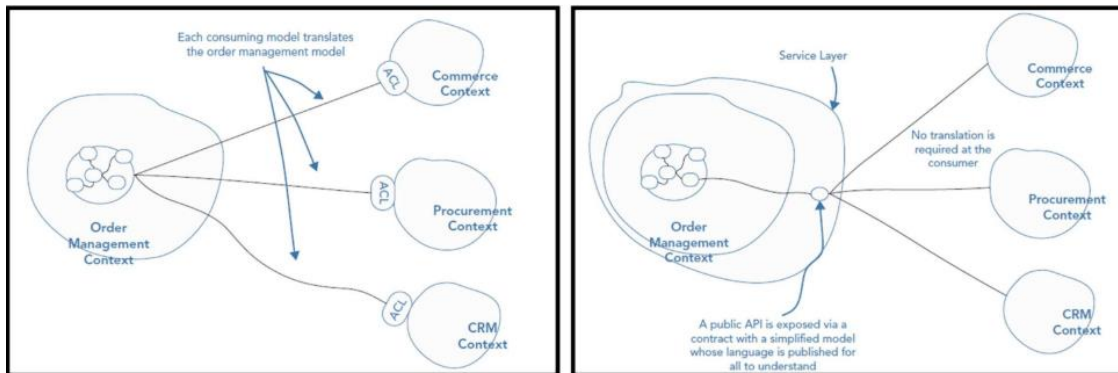
Benefit:

The ACL ensures the new system doesn't directly interact with the outdated structure, protecting it from being affected by the complexities of the legacy system.

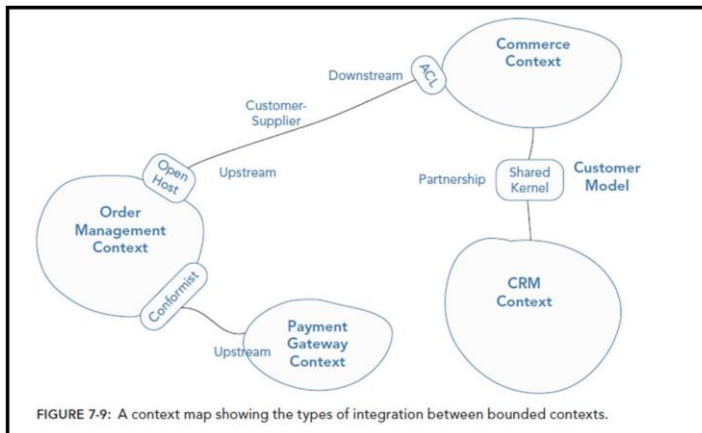
- **Shared Kernels:** Commonly shared logic between contexts.



- **Open Host Services (OHS):** Standardized and publicly available APIs (Application Programming Interface) provided by a **bounded context** to allow other systems or contexts to interact with it in a consistent, controlled way.



Communicating the Context Map



1. Upstream

- **A context that provides data or services consumed by other contexts (downstream).**

• Characteristics:

- It influences the downstream context.
- Changes made upstream can impact downstream systems.

• Example:

- A Product Catalog context (upstream) provides product details to an Order Management context (downstream).

2. Downstream

- **A context that depends on or consumes data or services provided by an upstream context.**

• Characteristics:

- It must adapt to the upstream's outputs or changes.

- **Example:**

- The Billing context (downstream) relies on the Order Management context (upstream) to provide finalized order details for invoicing.

3. Conformist (follower)

- **A downstream context that adopts the upstream context's model and language without applying isolation strategies.**

• Characteristics:

- Simplifies integration by aligning with the upstream's terms and design.
- Reduces effort but creates tight coupling, making it vulnerable to upstream changes.

• Example:

- A Customer Support context directly uses the User Management context's data model and terminology (e.g., "user roles," "user IDs") instead of defining its own.

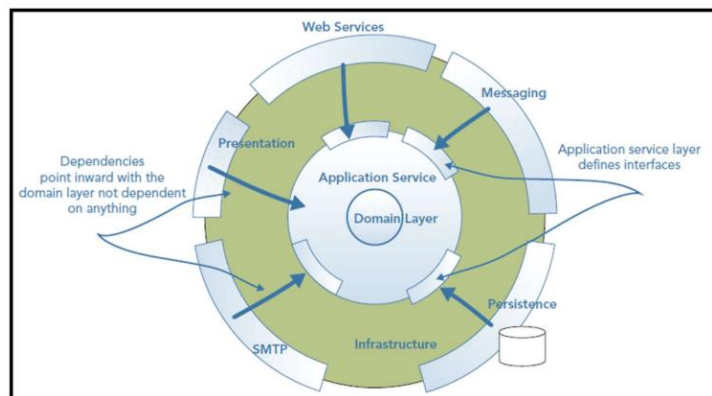
2. Application Architecture

Layered Architecture

- Organizes software into distinct layers to separate concerns:
 - Domain Layer:** The **core (center)** of the application where the business rules, logic, and processes reside. Focuses on solving the business problem, independent of technology or user interface.
 - Key Features:**
 - Represents the **domain model**, including entities, value objects, and aggregates.
 - Contains **business rules** that dictate how the domain operates.
 - Example:**
In an **e-commerce system**, the domain layer might validate if an order can be processed based on inventory, calculate discounts or taxes, ensure business rules, such as "orders above \$500 must be manually reviewed."
 - Application Layer:** Acts as a **mediator (between two layers)** between the domain layer and the external systems (like the user interface or other services). Coordinates **application-specific workflows** without implementing business logic itself.
 - Key Features:**
 - Orchestrates **use cases** or application flows by invoking the domain layer.
 - Manages **communication** with external layers (like infrastructure).
 - Example:**
In the same **e-commerce system**, the application layer might:
 - Trigger a "place order" use case by calling domain services to validate the order and then passing it to the infrastructure layer to save in the database.
 - Send confirmation emails by coordinating with the infrastructure layer.
 - Infrastructure Layer:** The layer responsible for handling **technical details** such as data storage, messaging, or external API communication. Supports the application and domain layers by providing access to external systems and frameworks.
 - Key Features:**
 - Contains implementations for **databases, message brokers, and APIs**.
 - Focuses on non-business concerns like logging, caching, or persistence.
 - Example:**
In the **e-commerce system**, the infrastructure layer might:
 - Save order details to a database.
 - Call a payment gateway API to process payments.
 - Log order processing events to a monitoring system.

SMTP: Simple Mail Transfer Protocol

Dependency Inversion within a layered architecture



- Ensures layers depend on abstractions rather than concrete implementations.
- Example: A service depending on an interface, not a specific database.

Domain vs. Application Logic

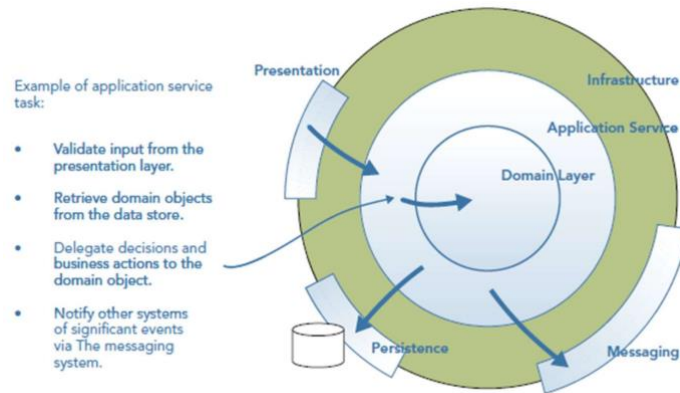


FIGURE 8-9: Application logic versus domain logic.

- **Domain Logic:** Core business rules (e.g., calculating interest rates).
- **Application Logic:** Use-case-specific tasks (e.g., triggering notifications).

3. Key Principles for Teams Starting with DDD

The tactical patterns of DDD can guide you toward creating an effective domain model; however, this area of DDD is evolving, and the implementation details have been overemphasized (gave more importance than it deserves). The patterns may have value, but this is not where the value of DDD lies.

- **Collaboration is Crucial:** Domain experts and developers must work together.
- **A Context is Everything;** context and isolation retain the integrity of your code. It reduces cognitive load and makes a model specific.
- **Domain Experts:** You need access to a domain expert. Teams can't reveal deeper insights without them.
- **Use CRUD for Bounded Contexts** with low complexity. You are not a bad programmer if you don't have a domain model.
- **Bounded Context and the Ubiquitous Language** are the foundation of DDD.
- **DDD** is about the process of learning, refining, experimenting, and exploring in the quest to discover a useful model in collaboration.
- **Don't sell DDD as a silver bullet (one-size-fits-all or easy fix).** Focus on the alignment with the business and learning more about the domain you are building software for.
- **Decompose the problem space** and focus on the core domain. All interesting conversations will happen here. This is where you apply the principles of DDD to maximize value and where you should apply the most effort.
- **Focus on the Domain:** Avoid over-engineering technical aspects.
- **Ubiquitous Language:** Use shared terminology that reflects the domain.
- **Bounded Contexts:** Clearly define boundaries to isolate models.
- **Before modeling a solution, Capture the landscape:** Understand other models and contexts in play, their ownership, relationships, and data exchanges.
- **Focus on feature scenarios:** Build a model starting with the most risky or complex features, using domain expert time wisely (avoid simple data management).
- **Protect yourself in legacy (a system or component built with older, possibly outdated technologies) environments:** Safeguard from external code, enforce boundaries, and carve out areas for new functionality without trying to clean up everything.
- **Constant integration and refinement:** Challenge your model by exploring new ideas and solutions. Aim for at least three useful models.
- **Simplify and delay decisions:** Avoid assumptions, keep things simple, and wait for complexity to guide design decisions, refactoring as needed.
- **Modeling is a team activity:** It should happen whenever the team is stuck or uncertain, not as a predefined project step.
- **Model and language evolve together:** Ensure the model is easily communicable and adaptable for future evolution.

Quiz Insights

Explanation of Answers:

1. **Modeling sessions** should be timeboxed, but the team should revisit and refine the model as needed.
2. **Primary Purpose** of A context map clarifies boundaries and team communication.
3. A **"Big Ball of Mud"** refers to a disorganized, monolithic codebase.
4. An **ACL** isolates and translates between contexts, protecting the domain model. Doesn't translate ubiquitous language
5. **The BIAN organization** is a collaborative ecosystem of banks, technology providers, and academics
6. **Open Host Services** provide standardized APIs for downstream integration.

Topic 10: Introducing the Enterprise Architecture

Key Themes:

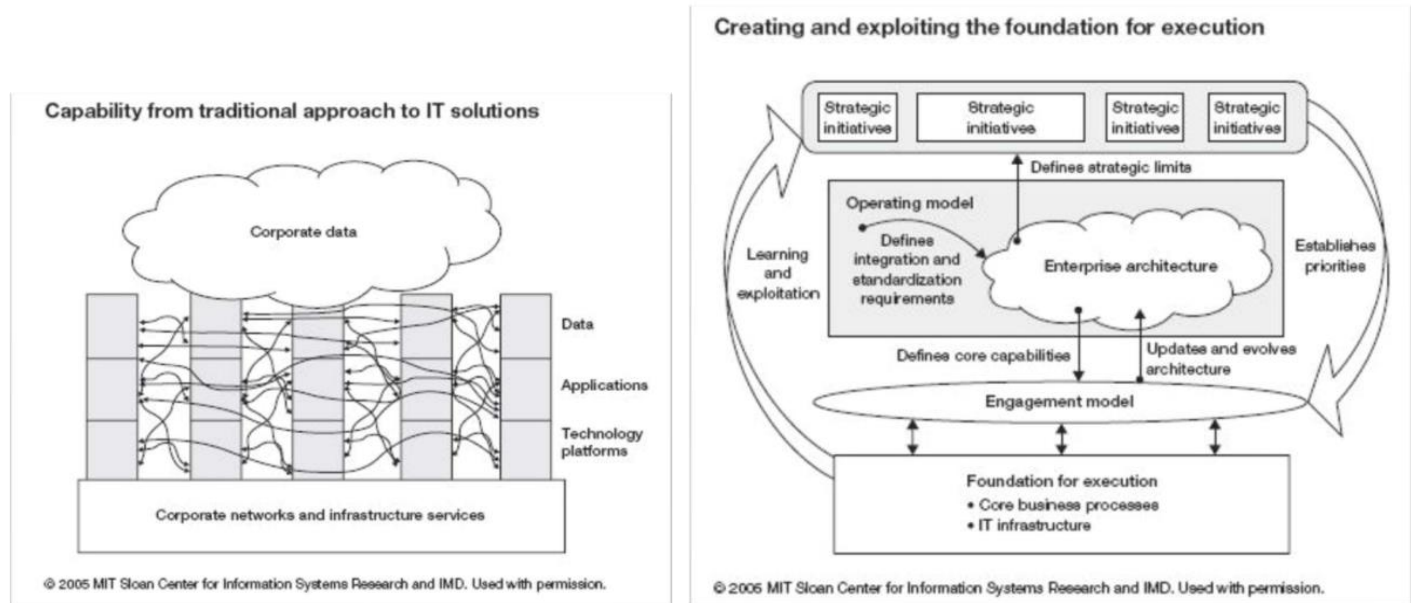
- Foundation for Execution
- Operating Model
- Implementing the Operating Model via Enterprise Architecture

Keywords for Quick Reference:

- **Foundation for Execution:** Search for topics related to IT infrastructure, routine process digitization, and strategic focus.
- **Operating Model:** Look for integration, standardization, and process alignment.
- **Enterprise Architecture:** Use for questions on blueprinting business processes, core diagrams, and levels of architecture.

Foundation for Execution

- **Definition:** A foundation for execution involves IT infrastructure and digitized business processes that automate an organization's core capabilities. It is essential for the organization to focus on high-order activities (**such as serving customers and responding to new business opportunities**) by streamlining routine processes.
- **Benefits:**
 - **Reliability and Predictability:** Helps ensure processes are executed correctly, minimizing errors.
 - **Management Focus:** Frees up management to concentrate on strategic activities rather than mundane (ordinary) tasks.
- **Paradox of Digitization:**
 - While digitizing core processes reduces their flexibility, it increases the overall agility of the organization. This mirrors the analogy of an athlete, where specialized capabilities enhance overall performance despite their rigidity.



Operating Model

- **Reliance on Strategy:** Most organizations use strategy to guide IT investments, aiming to align IT and business processes with the overall business strategy.
- **Elusiveness (difficult to describe) of Alignment:** Achieving alignment between business and IT can be challenging due to the complexity of strategies.
- **Multifaceted (having many aspects) Strategies:** Business strategies involve decisions on market competition, positioning, and capability development.
- **Shifting Priorities:** Strategic priorities often change in response to competitor actions or new opportunities.
- **Lack of Clear Direction:** Strategy typically does not provide a clear path for developing stable IT infrastructure and business process capabilities.
- **Definition:** An operating model describes the necessary level of business process integration and standardization to deliver services efficiently. By providing a more stable and actionable view of the organization than strategy, the operating model drives the design of the foundation for execution
- **Dimensions:**

Executives need to recognize standardization and integration as two separate decisions

- **Business Process Standardization:**
 - Ensures processes are executed consistently across units regardless of who is performing the process or where it is completed, reducing variability and improving efficiency.
 - The result of standardization (a reduction in variability) can be dramatic increases in throughput and efficiency
 - Limits local innovation and may require replacing perfectly good (and occasionally superior) existing systems and processes. This can be politically difficult and expensive
- **Business Process Integration:**
 - Enable end-to-end transaction processing or across processes, allow the organization to present a single face to customers
 - Involves linking organizational units through shared data, enhancing transparency, coordination, and agility and allow changes in one part of the business to alert other parts of actions they need to take.
 - Integration can also speed up the overall flow of information and transactions through an organization
 - Challenges include agreeing on standard definitions and formats for data.
 - For business units to share customer information, they must agree on its format. Similarly, they must share a common definition for terms like sale, which can be said to occur when a contract is signed, when money is paid, or when product is delivered. These can be difficult, time-consuming decisions

Characteristics of four operating models:

Business process integration	High	Coordination • Shared customers, products, or suppliers • Impact on other business unit transactions • Operationally unique business units or functions • Autonomous business management • Business unit control over business process design • Shared customer/supplier/product data • Consensus processes for designing IT infrastructure services; IT application decisions made in business units	Unification • Customers and suppliers may be local or global • Globally integrated business processes often with support of enterprise systems • Business units with similar or overlapping operations • Centralized management often applying functional/process/business unit matrices • High-level process owners design standardized processes • Centrally mandated databases • IT decisions made centrally
	Low	Diversification • Few, if any, shared customers or suppliers • Independent transactions • Operationally unique business units • Autonomous business management • Business unit control over business process design • Few data standards across business units • Most IT decisions made within business units	Replication • Few, if any, shared customers • Independent transactions aggregated at a high level • Operationally similar business units • Autonomous business unit leaders with limited discretion over processes • Centralized (or federal) control over business process design • Standardized data definitions but data locally owned with some aggregation at corporate • Centrally mandated IT services
		Low	High
		Business process standardization	

© 2005 MIT Sloan Center for Information Systems Research. Used with permission.

Figure 2-1 presents a two-dimensional model with four quadrants that classify companies based on their levels of business process integration and standardization. Each quadrant represents a distinct operating model, which helps companies clarify how they deliver goods and services to customers. The models are:

1. **Diversification** (low standardization, low integration): Business units operate independently, offering distinct products and services with limited central control. Synergies are achieved through shared services and brand recognition.
2. **Coordination** (low standardization, high integration): Business units share customers, products, or suppliers, benefiting from integrated processes like cross-selling and supply chain transparency while maintaining unique operations.
3. **Replication** (high standardization, low integration): Business units operate independently but follow standardized, repeatable processes for efficiency and scalability, such as franchise models.

4. **Unification** (high standardization, high integration): Business units are tightly integrated and standardized, sharing transaction data and processes to maximize efficiency and customer service.

To determine the appropriate operating model, companies evaluate their integration and standardization needs by asking how dependent business units are on shared data and whether uniform operations are beneficial. The chosen model guides decisions on IT systems, business unit autonomy, and process design.

Implementing the Operating Model via Enterprise Architecture

- **Operating Model Expectations:** Defines general expectations for business process integration and standardization across units.
- **Enterprise Architecture Overview:**
 - Delineates (describes) key processes, systems, and data central to the organization's operations.
 - Guides the digitization efforts to build the foundation for execution.
 - Organize business processes and IT infrastructure to meet integration and standardization needs of the operating model.
- **Effective Implementation:** The key to **effective enterprise architecture** is to identify the processes, data, technologies, and customer interfaces that take the operating model from vision to reality

High-level enterprise architecture fosters a common understanding of organizational operations, requiring detailed alignment (convergence) of people, processes, and technology. Implementation necessitates four levels of architecture: business process, data/information, applications, and technology architecture.

- **Levels of Architecture:**
 - **Business Process Architecture:** The activities or tasks composing major business processes identified by the business process owners.
 - **Data/Information Architecture:** Shared data definitions.
 - **Applications Architecture:** Individual applications and their interfaces.
 - **Technology Architecture:** Infrastructure services and the technology standards they are built on.
- **Representation of Enterprise Architecture:** Unlike building blueprints, enterprise architecture is conveyed through principles, policies, and technology choices, which can be challenging for managers to grasp.
- **Core Diagram:** A simplified picture, known as the "core diagram," aids managers in debating and eventually understanding the organization's enterprise architecture.
- **Core Diagram Components:**
 - **Core Business Processes:** Small set of enterprise processes which define the stable set of organization-wide capabilities the organization needs to execute its operating model and respond to market opportunities
 - **Shared Data:** This includes data, such as customer files shared across the product lines of a full-service financial services institution or master supplier and item data, shared across business units of an organization instituting (starting) the global supply chain
 - **Key Linking and Automation Technologies:**
 - These technologies include "**middleware**," which facilitates the integration of various applications and access to shared data, along with **significant software packages** like **ERP** (Enterprise Resource Planning) systems.
 - Additionally, key technologies include portals that offer standardized access to systems and data, or customer interfaces that help differentiate an organization from its competitors. Electronic interfaces connecting key stakeholder groups such as employees, customers, partners, and suppliers also might appear on an enterprise architecture core diagram
 - **Key Customers:** show the major customer groups (e.g., channels or segments) served by the foundation for execution

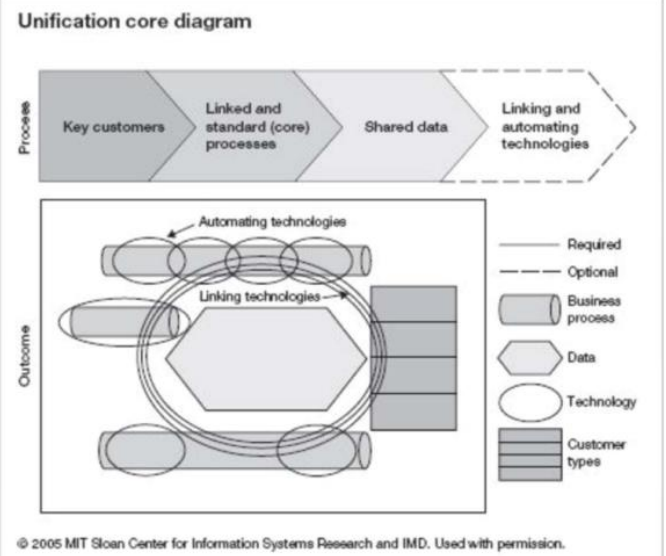
Diagrams:

Unification core diagram:

Figure 3-2 outlines the process for designing the enterprise architecture core diagram for a Unification company. This involves:

1. Identifying key customers (segments and channels).
2. Listing key processes to standardize and integrate.
3. Determining shared data needed for process integration and customer service.
4. Optionally including key technologies that automate or link processes (indicated with dashed outlines in the figure).

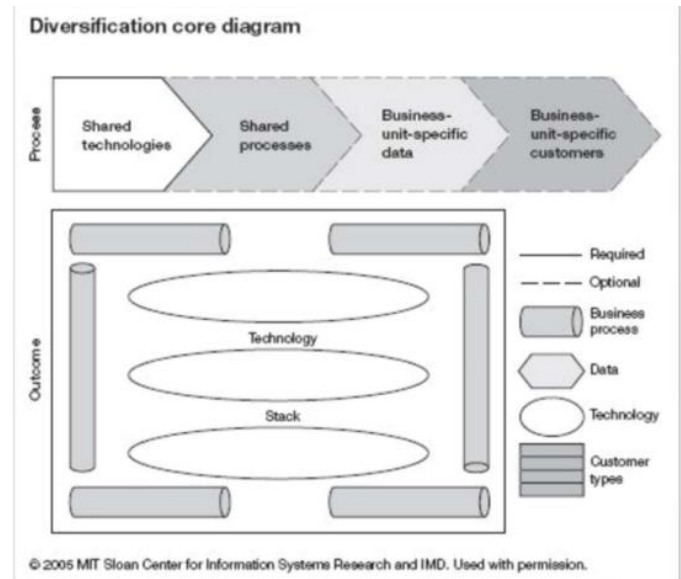
The bottom half of the figure illustrates a core diagram for a Unification company, showcasing a highly standardized and integrated system. It highlights standard processes accessing shared data to deliver products and services. Key technologies may be included if they are central to the company's management vision.



Diversification Core Diagram:

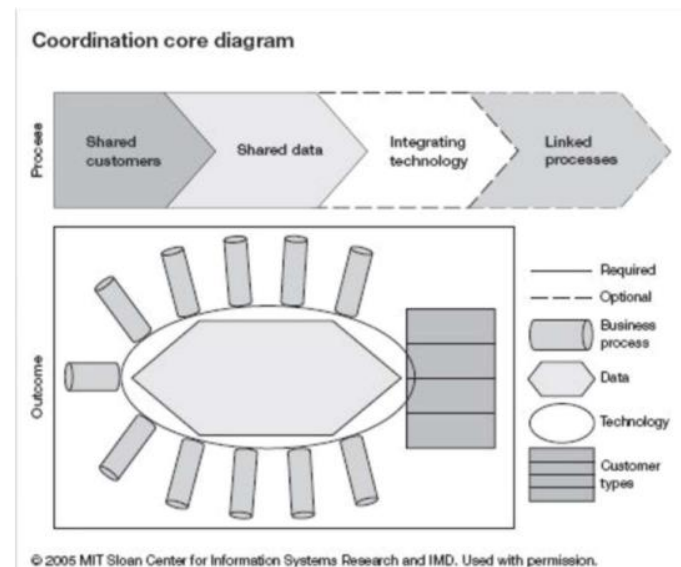
An extreme Diversification model lacks enterprise architecture entirely, with no synergies across businesses. More commonly, Diversification companies establish shared technologies to achieve economies of scale. These technologies such as data centers, telecommunications networks, vendor negotiations, and help desks, form the core of the enterprise architecture diagram. Some companies may also include limited shared processes or data, particularly for financial reporting, risk management, or compliance.

When designing a Diversification core diagram, prioritize shared technologies that deliver economies of scale or standardization. Other elements, like key customers, processes, and data, are included only if essential to the operating model. However, offering a unified interface to common customers is rare in Diversification companies.



Coordination Core Diagram:

The **Coordination model** focuses on delivering integrated services to key customer groups by sharing critical data across business units, enabling a unified customer experience. This approach is commonly adopted by companies with diverse product offerings, such as large financial institutions like MetLife. Coordination facilitates the integration of numerous products and processes without requiring standardization, allowing for flexibility and customization within business units.



Replication Core Diagram:

Replication operating models focus on standardized processes, which are central to their enterprise architecture core diagram. The diagram typically highlights key standardized processes and the technologies enabling them, as well as shared technologies linking these processes. Data is rarely included, as business units in Replication models do not usually share data.

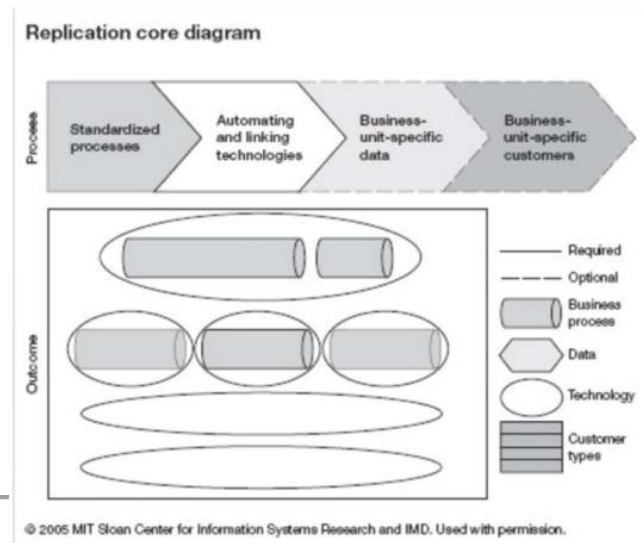
To enhance efficiency, Replication companies automate key processes and often develop reusable business modules. When designing a Replication core diagram:

1. Start with the key processes to standardize and replicate across business units.
2. Identify the technologies automating these processes.
3. Consider shared linking technologies, if applicable.

Key customers and shared data are generally not included, as business units handle these aspects independently.

Quiz Insights:

- **Enterprise Architecture's Role:** It helps align IT and business strategies by digitizing core processes and providing a shared understanding of operations.



Topic 11: Enterprise Architecture (EA) Practices

Learning Objectives

- Understand the comparison between Enterprise Architecture (EA) and City Planning.
- Learn the six types of EA artifacts and their relationships.
- Recognize and analyze the CSVLOD model.
- Discuss the interaction between business and IT in EA.
- Explore the EA Uncertainty Principle.
- Understand the three core processes of EA practice.

Keywords for Quick Reference

- **EA vs. City Planning:** For general comparisons.
- **CSVLOD Model:** Rules, structures, changes.
- **Artifact Types:** Considerations, Standards, Visions, Landscapes, Outlines, Designs.
- **EA Processes:** Strategic Planning, Initiative Delivery, Technology Optimization.
- **Dynamics:** Business-IT alignment, challenges.
- **Uncertainty Principle:** Evolving priorities, unpredictability.

1. EA vs. City Planning

Enterprise architecture	City planning	Commonalities between enterprise architecture and city planning
Business executives	City governors	Both business executives and city governors are interested in the long-term prosperity of their organizations or cities from the perspective of their ultimate value, but not necessarily aware of their technical Infrastructure
IT project teams	Construction project teams	Both IT project teams and construction project teams are responsible for completing their technical projects on time, but may be unaware of their long-term impact and ultimate value
Architects	City planners	Both architects and city planners are accountable for translating the long-term ultimate value perspective of business executives or city governors into the short-term technical perspective of IT project teams or construction project teams

Enterprise Architecture (EA) and city planning share several parallels:

- **Business Executives vs. City Governors:** Both focus on long-term prosperity without necessarily understanding technical infrastructure.
- **IT Project Teams vs. Construction Teams:** Both execute technical projects, unaware of their long-term strategic value.
- **Architects vs. City Planners:** Translate long-term visions into actionable technical projects.

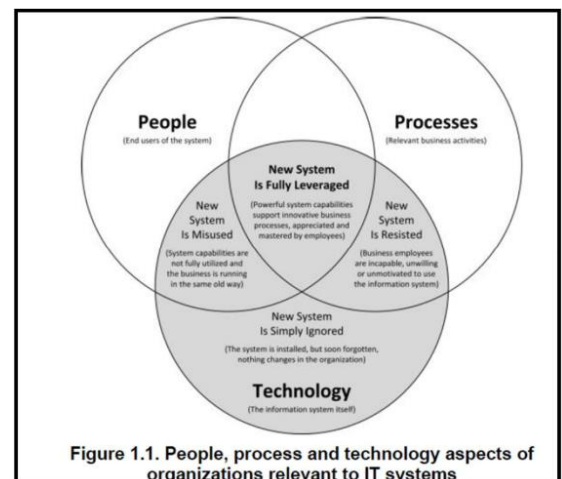
Individual Components:

1. People

- Represents the **end-users** or employees who interact with the IT system.
- Key responsibilities:
 - Learn and adapt to the system.
 - Use the system effectively in their daily tasks.
- Challenges:
 - Lack of training or skills.
 - Resistance to change or fear of job displacement.
- Example: Employees using a customer relationship management (CRM) system to track client interactions.

2. Processes

- Refers to the **business workflows and activities** that the IT system is designed to support or enhance.



- Key responsibilities:
 - Define clear objectives and use cases for the system.
 - Update or redesign workflows to utilize the system efficiently.
- Challenges:
 - Outdated or rigid processes that don't align with the system's capabilities.
 - Lack of process optimization or integration with the technology.
- Example: Automating manual invoicing processes through the new system.

3. Technology

- Refers to the **IT system** itself, including software, hardware, and infrastructure.
- Key responsibilities:
 - Provide the necessary tools and features to support business processes.
 - Be reliable, user-friendly, and scalable.
- Challenges:
 - Poor system design or functionality.
 - Lack of customization to suit organizational needs.
- Example: A well-designed ERP system that integrates all departments for better coordination.

Integration of Components:

People + Processes (Excluding Technology):

- When employees are skilled, and workflows are well-defined, but the technology is inadequate.
- Outcome: **New System Is Simply Ignored**
 - The system is underutilized or abandoned because it doesn't provide value or fit the organization's needs.

Processes + Technology (Excluding People):

- When the technology is advanced, and processes are optimized, but employees are unwilling or unable to adapt.
- Outcome: **New System Is Resisted**
 - Employees resist using the system due to poor training, lack of motivation, or fear of change.

People + Technology (Excluding Processes):

- When employees are trained, and the system is functional, but workflows remain outdated.
- Outcome: **New System Is Misused**
 - The system is used for old, inefficient processes, preventing true innovation.

People + Processes + Technology:

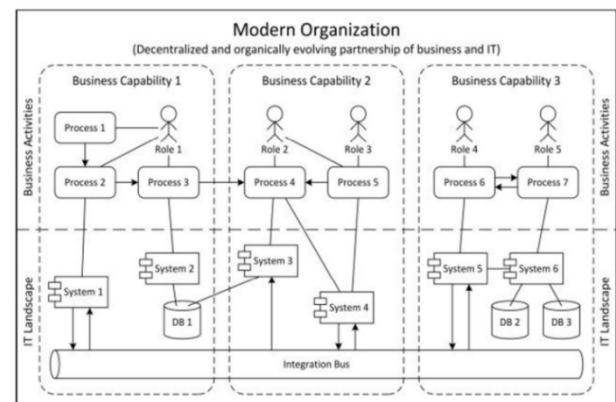
- When all three components are aligned and work together seamlessly.
- Outcome: **New System Is Fully Leveraged**
 - The system supports innovative workflows, employees are skilled and motivated, and the organization achieves maximum value from the IT system.

This ideal integration results in improved productivity, efficiency, and organizational success.

This diagram shows how a **modern organization** integrates **business activities** and the **IT landscape**:

Key Points:

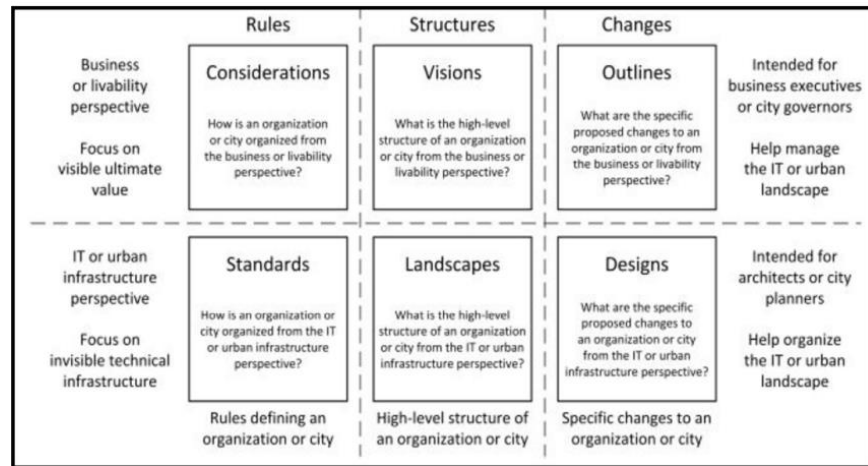
1. **Business Capabilities:**
 - Represent different organizational areas (e.g., Capability 1, 2, 3).
 - Include **Roles** (individuals) and **Processes** (workflows).
2. **IT Landscape:**
 - Includes **Systems** (e.g., System 1, System 2), **Databases**, and the **Integration Bus**.
 - The **Integration Bus** connects systems to ensure smooth communication and avoid silos.
3. **Interactions:**
 - Processes are supported by specific IT systems (e.g., Process 1 uses System 1).
 - Roles interact with processes to execute business activities.



Summary:

This structure reflects a **decentralized approach**, where business processes are mapped to IT systems, and the integration layer ensures collaboration and scalability across the organization.

2. Six Types of EA Artifacts (CSVLOD model)



This framework divides the planning and organization of a system (business, city, IT, or urban infrastructure) into **four perspectives** across two axes:

Axes:

- Horizontal Axis:**
 - Rules:** Define how the system is organized.
 - Structures:** Describe the high-level architecture.
 - Changes:** Focus on specific modifications or proposals.
- Vertical Axis:**
 - Business or Livability Perspective:** Focuses on visible value and outcomes for stakeholders.
 - IT or Urban Infrastructure Perspective:** Focuses on the underlying technical infrastructure.

Quadrants:

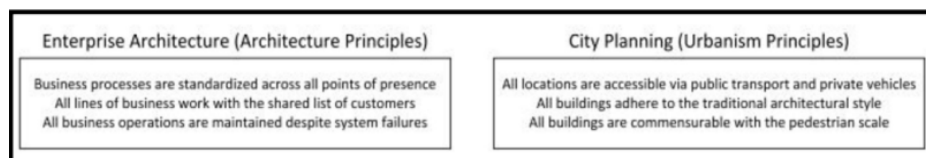
- Considerations:**
 - Rules** from a business/livability perspective.
 - Example: How a city or organization ensures livability or aligns with core values.
- Visions:**
 - Structures** from a business/livability perspective.
 - Example: High-level plans for livable spaces or business alignment.
- Outlines:**
 - Changes** from a business/livability perspective.
 - Example: Proposed initiatives to improve livability or business outcomes.
- Standards:**
 - Rules** from an IT/urban infrastructure perspective.
 - Example: Technical policies or protocols that define operations.
- Landscapes:**
 - Structures** from an IT/urban infrastructure perspective.
 - Example: Architectural diagrams for IT systems or urban infrastructure.
- Designs:**
 - Changes** from an IT/urban infrastructure perspective.
 - Example: Specific upgrades to IT systems or urban environments.

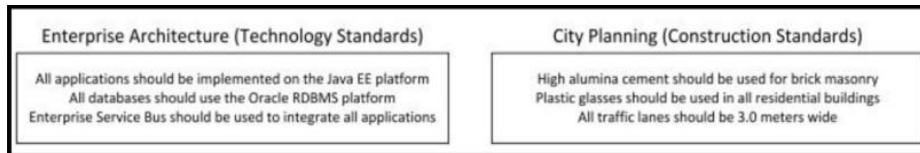
This framework helps different stakeholders (executives, architects, planners) manage and organize systems, balancing **visible outcomes** with **technical infrastructure** across **rules, structures, and changes**.

Key Terms:

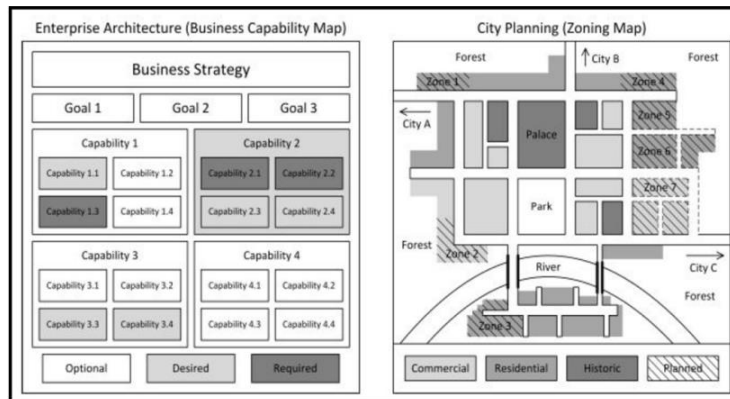
- Artifact:** Any document or diagram representing architectural elements.

Example of Considerations and Standards artifacts

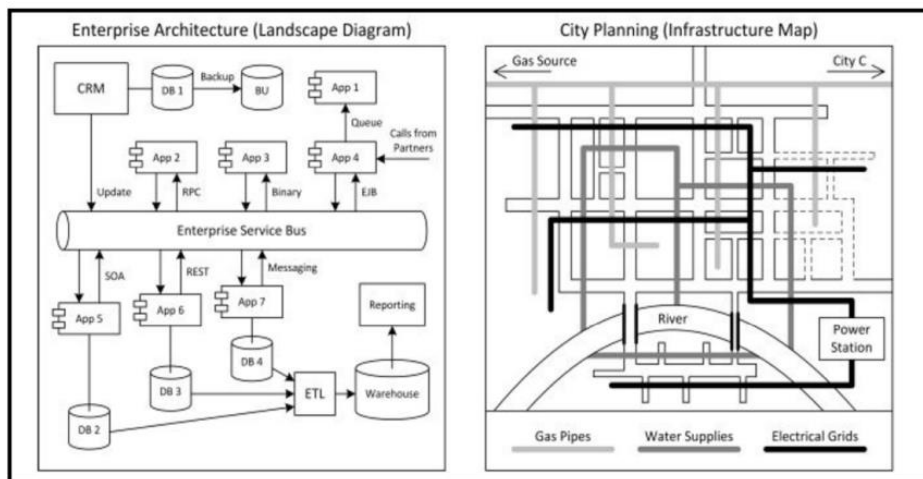




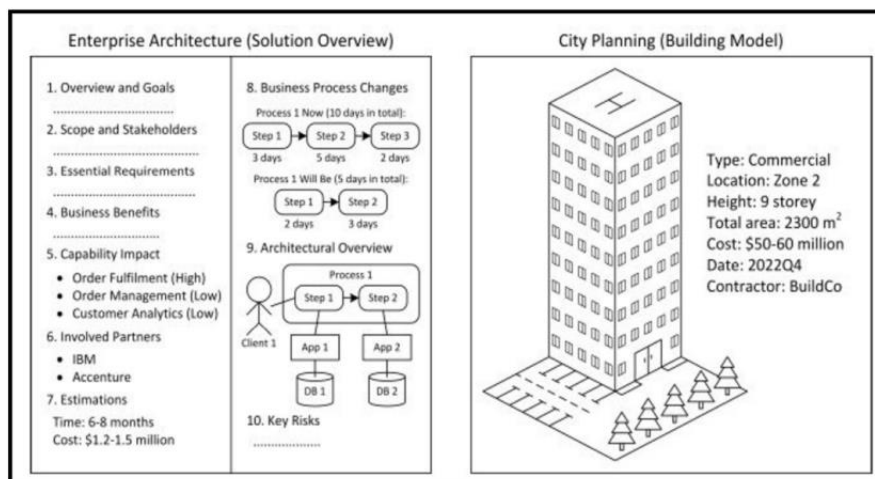
Example of Vision artifact



Example of Landscape artifact



Example of Outline artifact



Example of Design artifact

Key Concepts According to the reference

1. Lifecycle

- **Permanent Artifacts:**
 - **Definition:** Long-lived artifacts that exist for many years, evolving alongside the organization.
 - **Characteristics:**
 - Created once and updated as necessary to reflect ongoing organizational changes.
 - May be developed proactively or accumulated reactively over time.
 - Constantly used, continuously maintained, and occasionally discarded when they become irrelevant.
 - **Examples:**
 - **Considerations:** Architecture principles and urbanism principles.
 - **Visions:** Business capability maps and zoning maps.
 - **Standards:** Technology standards and construction standards.
 - **Landscapes:** Landscape diagrams and infrastructure maps.
 - **Temporary Artifacts:**
 - **Definition:** Short-lived artifacts that typically exist for several months or weeks.
 - **Characteristics:**
 - Created for specific purposes at particular moments.
 - Used as intended and then discarded or archived.
 - Due to their short lifespan, they usually do not require updates or maintenance.
 - **Examples:**
 - **Outlines:** Solution overviews and building models.
 - **Designs:** Solution designs and building blueprints.
-

2. Meaning

- **Decisions Artifacts:**
 - **Definition:** Artifacts that represent made planning decisions, reflecting formalized agreements between stakeholders regarding the desired future course of action.
 - **Characteristics:**
 - Have implications for the future and often imply specific organizational changes.
 - Developed or updated collaboratively by relevant stakeholders.
 - Inherently subjective and shaped by the interests of their stakeholders.
 - **Examples:**
 - **Considerations:** Architecture principles and urbanism principles.
 - **Visions:** Business capability maps and zoning maps.
 - **Outlines:** Solution overviews and building models.
 - **Standards:** Technology standards and construction standards.
 - **Designs:** Solution designs and building blueprints.
 - **Facts Artifacts:**
 - **Definition:** Artifacts that represent documented objective facts, reflecting the actual current situation in an organization.❏
 - **Characteristics:**
 - Do not imply any planning decisions and have no implications for the future.❏
 - May be developed or updated by specific actors without requiring broad stakeholder collaboration.❏
 - Based on acknowledged "hard" data and largely independent of specific people involved in their development.❏
 - **Examples:**
 - **Landscapes:** Landscape diagrams and infrastructure maps.❏
-

3. Duality

- **Definition:** The property of artifacts being relevant to two different audiences (**Strategic-Level Audience, Operational-Level Audience**) simultaneously, satisfying the information needs of both, and presented in a format appealing to both.
 - **Significance:**
 - Enables artifacts to serve as instruments of communication and partnership between diverse stakeholders involved in strategic decision-making and IT systems implementation.
 - Considered a fundamental mechanism underpinning an EA practice, facilitating effective collaboration among stakeholders.
-

1. Considerations

(Architecture principles and urbanism principles)

- **Who Uses Them?**
Business executive and architects or city governors and city planners.
 - **How Are They Used?**
Established collaboratively and then influence all planning decisions.
 - **Why Are They Used?**
improve overall conceptual consistency and mutual alignment.
 - **Key Details:**
 - **Dual:** Yes (used for both strategic and operational decisions)
 - **Meaning:** Decisions (helps in choosing consistent approaches)
 - **Lifecycle:** Permanent (these principles remain valid over time)
-

2. Visions

(Business capability maps and zoning maps)

- **Who Uses Them?**
Business executives and architects or city governors and city planners.
 - **How Are They Used?**
Developed collaboratively and then direct future investments and projects.
 - **Why Are They Used?**
To improve the strategic effectiveness of future investments.
 - **Key Details:**
 - **Dual:** Yes (guides both broad strategies and specific actions)
 - **Meaning:** Decisions (used for prioritizing and planning future work)
 - **Lifecycle:** Permanent (remains valid for the organization's long-term goals)
-

3. Outlines

(Solution overviews and building models)

- **Who Uses Them?**
Business executives and architects or city governors and planners.
 - **How Are They Used?**
Developed collaboratively and then inform project funding decisions.
 - **Why Are They Used?**
To improve cost-benefit ratio and efficiency of all investments.
 - **Key Details:**
 - **Dual:** Yes (supports both strategy and specific project-level funding decisions)
 - **Meaning:** Decisions (helps allocate funding and resources)
 - **Lifecycle:** Temporary (relevant only during the project's planning and funding phase)
-

4. Standards

(Technology standards and construction standards)

- **Who Uses Them?**
ONLY Architects or city planners.
 - **How Are They Used?**
Established and then influence all project designs as well as the landscape structure.
 - **Why Are They Used?**
To reduce complexity, risks and costs, and achieve homogeneity and compliance (Strong match).
 - **Key Details:**
 - **Dual:** No (focused only on operational compliance and quality)
 - **Meaning:** Decisions (provides guidance for consistency and quality)
 - **Lifecycle:** Permanent (standards rarely change and remain relevant over time)
-

5. Landscapes

(Landscape diagrams and infrastructure maps)

- **Who Uses Them?**
Only architects or city planners.
- **How Are They Used?**
Maintained current to support technical decision making and assist in project planning.

- **Why Are They Used?**
To rationalize infrastructure, reuse assets, and accelerate project planning.
- **Key Details:**
 - **Dual:** No (focused purely on representing factual, technical data)
 - **Meaning:** Facts (represents current infrastructure or design state)
 - **Lifecycle:** Permanent (updated but always relevant for planning)

6. Designs

(Solution designs and building blueprints)

- **Who Uses Them?**
Architects and IT project teams or city planners and construction project teams.
- **How Are They Used?**
Developed collaboratively and then guide the implementation of the project.
- **Why Are They Used?**
Improve quality of the project delivery according to requirements
- **Key Details:**
 - **Dual:** Yes
 - **Meaning:** Decisions (used to guide detailed project execution)
 - **Lifecycle:** Temporary (valid only during project implementation)

3. CSVLOD model:

Central Element:

- The central element of the CSVLOD model is the taxonomy (classification) for **Enterprise Architecture (EA) artifacts**.
- It provides a **classification scheme** to group diverse EA artifacts into six consistent categories.

Two Orthogonal Dimensions:

- **First Dimension:**
 - Classifies EA artifacts into **Rules, Structures, and Changes**.
 - Classifies based on what objects they describe

Artifacts	Rules	Structures	Changes
Describe	General global rules defining an organization or its divisions	High-level structures of an organization or its parts	Specific proposed incremental changes to an organization
Scope	Very broad, often relate to an entire organization	Broad, often cover large areas of an organization	Narrow, limited to separate IT initiatives or projects
Format	Often textual	Usually graphical	Mix of textual and graphical
Question	How do we work or want to work?	What approximately do we have or want to have?	What exactly are we going to change right now?
Lifecycle	Permanent, created once and then periodically updated	Permanent, created once and then continuously updated	Temporary, created for specific purposes and then discarded
Role	Basis for all other planning decisions	High-level "maps" facilitating decision making	Tactical plans of an organization
Purpose	Help achieve consistency and homogeneity of all planning decisions	Help understand what changes are desirable and how to implement them	Help plan separate changes in detail

- **Second Dimension:** Classifies EA artifacts into **Business-Focused** and **IT-Focused**.

Artifacts	Business-focused	IT-focused
Language	Technology-neutral business language	Technical IT-specific language
Domains	Business domain and often other relevant domains at a high level	Applications, data, integration, infrastructure, security domains and sometimes business domain
Format	Brief, intuitive, largely informal and include only the most essential information	Can be voluminous, formal, use strict notations and include comprehensive details
Stakeholders	Business leaders and architects	Architects and other IT specialists
Role	Communication interfaces between business and IT	Internal IT tools invisible to business
Purpose	Help business leaders manage IT	Help architects organize IT

4. Dynamics Between Business and IT

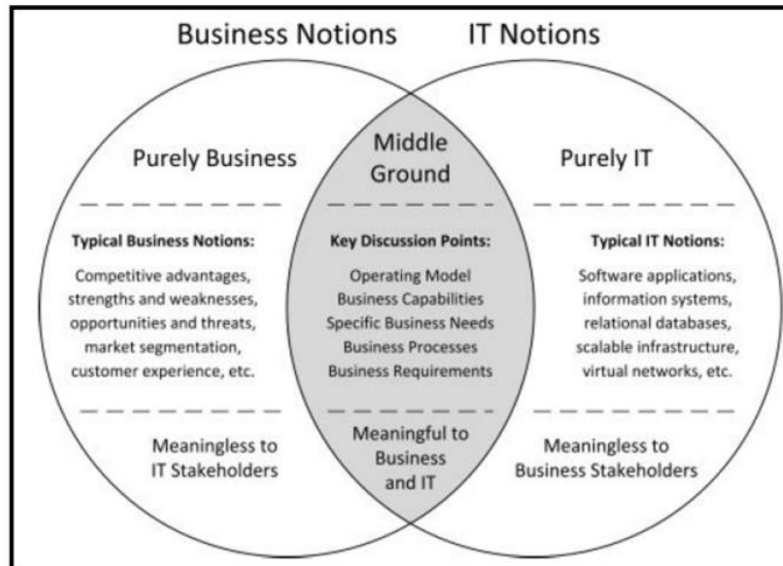
Problems with the Business Strategy as the Basis for Architectural Planning:

- **Vague Strategies:** Business strategies is often vague, unknown or merely absent.
- **Unclear Direction:** Business strategy rarely provides a clear direction for IT
- **Instability:** Business strategy is often unstable and frequently changes
- **Non-Reusable Systems:** Business strategy often requires strategy-specific, non-reusable IT Systems

Key Discussion Points Between Business and IT Stakeholders Global process standardization.

Refers to the organization's efforts to unify and streamline processes across all locations or departments
Ensures consistency, efficiency, and scalability in operations worldwide.

- **Operating Model:** Focuses on how the organization operates in terms of process standardization and **data sharing requirements**:
 - **Global Process Standardization:** Establishing uniform processes across all areas of the business.
 - **Data Sharing Requirements:** Defining how data is shared and integrated across systems, departments, or geographies.
- **Business Capabilities:** Identifies the specific capabilities (functions or competencies) of the organization that need improvement to meet business objectives.
 - Example: Enhancing supply chain management, customer service, or financial reporting.
- **Specific Business Needs:** Highlights particular pain points or challenges that the organization wants to address with IT solutions.
 - Example: Automating manual workflows or improving decision-making with data analytics.
- **Business Processes:** Refers to **high-level process change requirements** that IT solutions should support.
 - Example: Redesigning the procurement process or implementing new workflows for customer onboarding.
- **Business Requirements:** Focuses on **detailed, specific requirements** for IT systems to meet business needs.
 - Example: A CRM system must allow real-time customer data updates or a financial system must comply with specific regulatory standards.

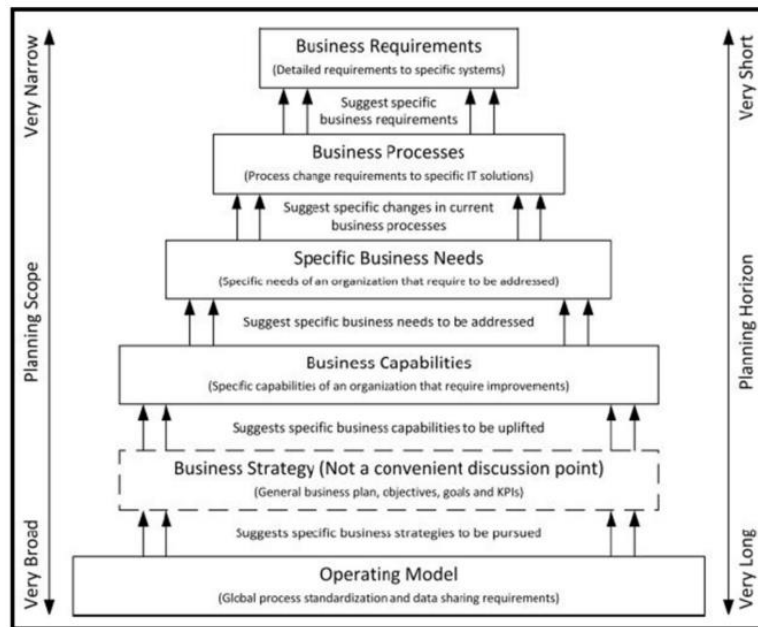


1. Two Notion Groups:

- **Business Notions:**
 - Refers to concepts and elements relevant to business stakeholders.
 - Includes areas like competitive advantages, market opportunities, customer experience, strengths, and weaknesses.
 - These notions are **meaningless to IT stakeholders**, as they are typically outside their technical focus.
- **IT Notions:**
 - Refers to concepts and elements relevant to IT stakeholders.
 - Includes areas like software applications, relational databases, scalable infrastructure, and virtual networks.
 - These notions are **meaningless to business stakeholders**, as they are too technical or operational.

2. The Middle Ground (Overlap):

- This is where **Business and IT align and collaborate** on shared priorities.
- Contains **key discussion points** that are meaningful to both groups, helping bridge the gap between their differing focuses.
- **Key discussion points include:**
 - **Operating Model:** How processes and data are standardized and shared globally.
 - **Business Capabilities:** Organizational functions needing improvement.
 - **Specific Business Needs:** Problems requiring IT solutions.
 - **Business Processes:** High-level process changes for IT support.
 - **Business Requirements:** Detailed system requirements to address specific needs.



This diagram represents a **hierarchy of business discussion points** between stakeholders, organized by **planning scope** (broad to narrow) and **planning horizon** (long-term to short-term)

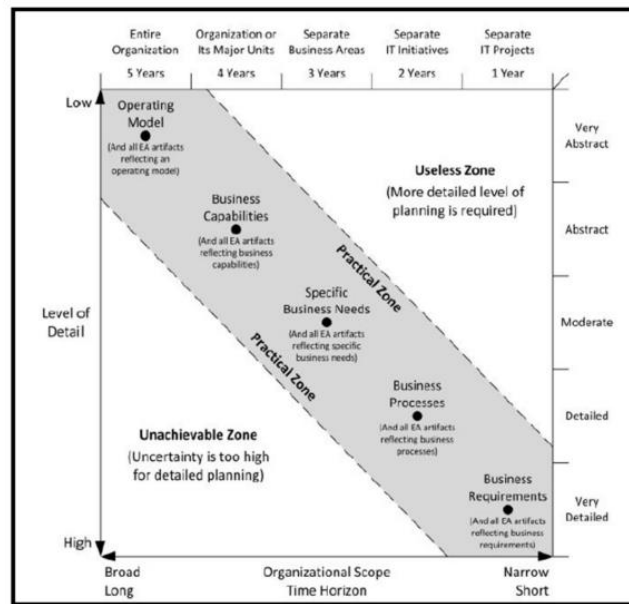
1. Layers of the Hierarchy (from bottom to top):

- **Operating Model:**
 - Focuses on global process standardization and data-sharing requirements.
 - Broadest planning scope with the longest planning horizon.
 - Provides a foundation for all other layers.
- **Business Strategy (Not a Convenient Discussion Point):**
 - Includes general business plans, objectives, goals, and KPIs (key performance indicators).
 - While important, it is not considered a "practical" discussion point between business and IT because of its abstract, high-level nature.
 - Suggests specific strategies to be pursued.
- **Business Capabilities:**
 - Identifies specific organizational functions that need to be improved.
 - Links strategy with actionable goals.
 - Suggests areas where capabilities can be uplifted to achieve strategic objectives.
- **Specific Business Needs:**
 - Highlights specific problems or requirements within the organization.
 - Suggests targeted issues that need to be addressed through IT or process improvements.
- **Business Processes:**
 - Focuses on high-level process change requirements for IT solutions.
 - Suggests changes to current processes to meet specific business needs.
- **Business Requirements:**
 - Provides detailed requirements for specific IT systems.
 - Narrowest planning scope with the shortest planning horizon.
 - Suggests system-level changes to support process improvements.

2. Key Concepts in the Diagram:

- **Planning Scope (Y-axis, Left):**
 - Indicates how broad or narrow the focus of the discussion point is.
 - Broader focus (e.g., Operating Model) encompasses organizational-wide concerns, while narrower focus (e.g., Business Requirements) deals with specific systems or solutions.
- **Planning Horizon (Y-axis, Right):**
 - Indicates the timeframe for planning and implementation.
 - Longer planning horizons (e.g., Operating Model) deal with strategic, long-term goals.
 - Shorter planning horizons (e.g., Business Requirements) deal with immediate, actionable tasks.

5. EA Uncertainty Principle



Key Elements of the Diagram:

1. **Axes of the Chart:**
 - **X-Axis (Organizational Scope and Time Horizon):**
 - **Broad, Long-Term Scope:** Entire organization with a 5-year horizon.
 - **Narrow, Short-Term Scope:** Separate IT projects with a 1-year horizon.
 - **Y-Axis (Level of Detail):**
 - **Low Detail:** Very abstract, strategic level.
 - **High Detail:** Very detailed, operational level.
2. **Zones of the Diagram:**
 - **Practical Zone:**
 - The middle diagonal shaded area where planning is feasible and meaningful.
 - This zone balances scope, time horizon, and level of detail to ensure that the plans are actionable and realistic.
 - **Unachievable Zone:**
 - Bottom-left area (high detail for long-term, broad scope).
 - Planning here is unrealistic because uncertainty makes detailed predictions impossible.
 - **Useless Zone:**
 - Top-right area (low detail for short-term, narrow scope).
 - Planning here is too abstract to provide actionable insights.
3. **Granularity of EA Planning:**
 - **Operating Model (Low Detail, Long-Term):**
 - High-level representation of how the organization operates (e.g., strategy and capabilities).
 - **Business Capabilities:**
 - Medium abstraction focusing on key business functions.
 - **Specific Business Needs:**
 - Focused planning addressing narrower, time-bound needs.
 - **Business Processes:**
 - Detailed plans for specific processes.
 - **Business Requirements (High Detail, Short-Term):**
 - Highly detailed planning, suitable for immediate and narrow-scope projects.

EA Uncertainty Principle (Key Takeaways):

1. **Trade-offs Between Detail, Scope, and Time Horizon:**
 - Detailed planning is only effective for short-term, narrow-scope efforts.
 - Broad, long-term planning must remain abstract to account for uncertainty.
2. **Unachievable Zone:**
 - It's impractical to create highly detailed plans for a broad organizational scope over long periods due to unpredictability.
3. **Useless Zone:**
 - High-level abstractions for short-term, narrow projects lack value because they don't provide sufficient actionable detail.
4. **Practical Zone:**
 - The sweet spot where planning is realistic and useful, balancing abstraction with feasibility.

5. Adaptability of EA Artifacts:

- Different levels of detail (from operating models to business requirements) correspond to varying scopes and time horizons. Plans should adjust to match these dimensions to remain effective.

6. Three Core EA Processes

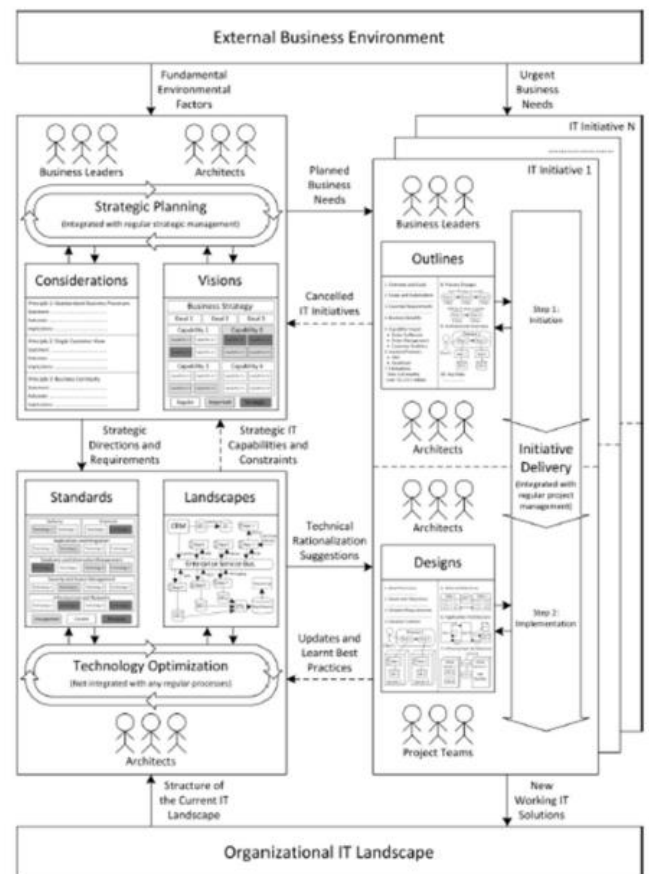
Three distinct high-level EA-related processes with different goals, participants and outcomes, These three processes are carried out simultaneously by different actors and interrelated with each other and they revolve around the six general types of EA artifacts described by the CSVLOD model:

a. Strategic Planning

- Translates relevant fundamental factors of the external business environment into more specific Considerations and Visions providing the general rules and directions for business and IT
- This process can be best summarized as strategy-to-portfolio
- The Considerations dimension covers:
 - What role should IT play in an organization?
 - Which IT capabilities should be provided organization-wide?
 - What IT-related policies should be complied with?
 - What level of business continuity and security is required?
 - Which IT-based business innovations should be adopted?
- The Vision dimension covers:
 - What should IT deliver to the business in the long run?
 - Where should future IT investments go?
 - What types of IT investments should be made?
 - When should IT investments be made?
 - In what sequence should IT investments be made?

b. Initiative Delivery

- Translates specific business needs, or more rarely specific technical needs, into tangible (touchable) IT solutions.
- Organizations typically have multiple instances of the Initiative delivery process (i.e. multiple separate IT initiatives) running in parallel and delivering different IT solutions simultaneously
- The Initiative Delivery process is tightly integrated with regular project management activities (e.g. scoping, estimating, scheduling, resourcing, monitoring, etc.) and follows a typical step-wise project delivery lifecycle with several sequential phases and control gates, e.g. scope, evaluate, plan, build, test and deploy
- regardless of organization-specific phases and gates of the project lifecycle, the Initiative Delivery process consists of two inherent consecutive steps: **initiation** and **implementation**.
- For large IT initiatives, some iterative features may be required in the process.



Initiation Step

- **Purpose:**
 - Formulates the goals and objectives of the IT initiative.
 - Generates potential implementation options, evaluates their pros and cons, and selects the most appropriate option.
 - Secures approval for the proposed high-level IT solution from all relevant business stakeholders (often referred to as a **feasibility study**).
- **Role of Outlines:**
 - Outlines (from the CSVLOD model) are critical for effective communication during the initiation step.
- **Key Questions Covered by Outlines:**
 - What do the proposed IT solutions look like conceptually?
 - How will the proposed IT solutions modify established business processes?
 - What is the immediate and long-term business value of the proposed IT solutions?
 - What is the overall business impact of the proposed IT solutions?
 - What is the cost of the proposed IT solutions, and when can they be delivered?

Implementation Step

- **Purpose:**
 - Involves architects, IT project teams, and business representatives to implement the approved solution.

- **Role of Designs:**
 - Designs (from the CSVLOD model) are key EA artifacts enabling discussions between architects, IT project teams and some business representatives during the implementation step.
- **Key Questions Covered by Designs:**
 - What business requirements should be addressed?
 - What new software should be developed or installed?
 - Which data types and entities should be used?
 - What servers and hardware should be deployed?
 - How exactly should new IT systems interact with the existing systems?

c. Technology Optimization:

Constant delivery of new IT solutions, even if these solutions are perfectly aligned with the business strategy, inevitably complicates the IT landscape, gradually tangles the interconnections between various systems, multiplies IT assets, proliferates (rapidly grow) supported technologies, introduces legacy and redundancy. These adverse side effects inflate the IT budget, reduce the strategic agility and flexibility of IT, increase IT-related risks and eventually create substantial problems for organizations in the long run. Technology Optimization is intended to reduce those negative effects

- **Purpose:**
 - Technology Optimization aims to untangle, simplify, and rationalize the organizational IT landscape and technology portfolio to reduce the negative effects of constant delivery of new IT solutions.
 - This process is summarized as "**structure-to-rationalization**".
 - **Challenges Addressed:**
 - Complication of the IT landscape.
 - Tangled interconnections between systems.
 - Multiplication of IT assets and supported technologies.
 - Introduction of legacy systems, redundancy, and inefficiencies.
 - Increased IT budget, reduced agility, higher risks, and long-term organizational challenges.
 - **Key Artifacts (CSVLOD Model):**
 - **Standards and Landscapes** are crucial to Technology Optimization.
 - **Standards Focus Areas:**
 - What technologies and vendor products are used?
 - What approaches and best practices are adopted?
 - Which technologies/products are redundant or fulfill similar purposes?
 - Which technologies/products cause problems?
 - Do current technologies and approaches meet business needs?
 - **Landscapes Focus Areas:**
 - What IT assets are maintained by an organization?
 - Which IT assets are not actively used or provide duplicated functionality?
 - Which IT assets are no longer supported by their vendors?
 - Which IT assets may cause problems in the future?
 - Are the existing IT assets adequate for the organizational business needs?
-

Topic 12: An Introduction to Governance

Key Themes

- Characteristics of a company.
- Types of companies and shareholding categories.
- Issues in corporate governance.
- Relationship between governance and corporate performance.
- Strengths and weaknesses of family-owned businesses.

Keywords

- Corporate governance principles.
- Shareholding patterns.
- Board composition and roles.
- Family business governance.
- Perpetual succession.

Characteristics of a Company

1. **Incorporated Association**
 - Created when registered under the Companies Act.
 - Gains recognition as a separate legal entity distinct from its members.
2. **Artificial Legal Person**
 - The company exists in the eyes of the law and cannot act on its own
 - Has to act through a board of directors elected by the shareholder
3. **Separate Legal Entity**
 - A company has a legal distinct entity and is independent of its members
 - The property of the company is to be used for the benefit of the company and not for the personal benefit of the shareholder
4. **Perpetual Succession**
 - The company remains regardless of changes in membership (death, exit).
 - Law creates it and law alone can dissolve it
5. **Common Seal**
 - The law has provided for the use of common seal, Serves as a substitute for company's official signature, with its name engraved.
6. **Separation of Ownership and Management**
 - Actual control and management is delegated by the shareholders to their elected representatives, known as directors

Types of Companies

Classification can be based on:

- Mode of incorporation
- Number of members
- Control
- Ownership of companies

By Ownership

1. **Proprietary (Sole Proprietorship)**
 - Owned and managed by one individual. (proprietor or sole trader)
 - Full risk and profit responsibility.
 - Contributes capital (providing money or assets to fund the establishment or growth of a business.)
 2. **Partnership**
 - Partners collectively own and manage the firm, sharing risks and profits. (Called individually as partner and collectively as firm)
 - Produce more capital and reduce individual risk
 3. **Joint Stock Company**
 - A large part of business all over the world assumed joint stock company format in term of volume and value.
 - Characteristics like separate entity, separation of ownership and control.
 - Combines ownership (shareholders) and professional management.
 - Allows risk diversification and share transferability of right.
 4. **Co-operative**
 - Voluntary association of persons aiming to achieve some common purpose, through an economic enterprise.
 - Work at their own risk with resources which all members contribute
-

Shareholding Categories

Shareholding pattern of a company shows how its shares are split among the entities that make up its owners. There are two main groups:

1. **The Promoter and The Promoter Group**
 - Founders or initial investors holding a controlling stake.
2. **Public Shareholding**
 - Shares distributed among the general public, enhancing liquidity.

Benefits of Shareholding

- Access to capital and investment diversification (Investors can diversify their investment portfolios by owning shares in multiple companies).
- According to liquidity demands, shares can be easily bought or sold
- Shareholders have ownership rights in the company, including voting rights on key decisions

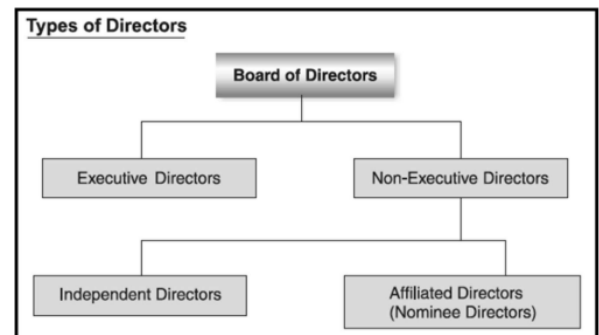
Issues in Corporate Governance

Corporate Governance is a means to an end, the end being long-term shareholder value, and more importantly, stakeholder value.

Thus, all authorities on the subject are one in recognizing the need for good corporate governance practices to achieve the end for which corporates are formed.

They identify some governance issues being crucial and critical to achieve these objectives.

1. **Distinguishing Board and Management Roles**
 - The board of a listed company has the following functions:
 1. To Select, decide the remuneration and evaluate on a regular basis, and when necessary, change the CEO
 2. To oversee (not directly, but indirectly) the conduct of the company's business to evaluate whether or not it is being correctly managed
 3. To review and, where necessary, approve the company's financial objectives and major corporate plans and objectives
 4. To render advice top management including the board of directors
 5. To identify and recommend candidates to the shareholders for electing them to the board of directors
 6. To review the adequacy of systems to comply with all applicable laws and regulations
 7. To perform all other functions required by law
2. **Composition of the Board**
 - The **board of directors** is a committee elected by shareholders to oversee the company's policies and strategic direction.
 - Some boards include **full-time functional directors** who manage specific branches of the company's operations.
 - The **composition of the board** varies in number and type of directors, evolving over time to reflect changing corporate governance needs.
- **Board of Directors**
 - Comprises two main categories: **Executive Directors** and **Non-Executive Directors**.
- **Executive Directors**
 - Actively involved in the day-to-day operations and management of the company.
 - Often hold senior roles, such as CEO or COO, alongside their board responsibilities.
- **Non-Executive Directors**
 - Do not participate in daily management but provide oversight and guidance.
 - Serve as independent advisors to ensure balanced decision-making.
 - **Types of Non-Executive Directors:**
 - **Independent Directors:**
 - Have no material relationship with the company, ensuring impartiality in governance.
 - Act as a check on management to protect shareholders' interests.
 - **Affiliated (Nominee) Directors:**
 - Represent the interests of specific stakeholders, such as major shareholders or investors.
 - May have ties to the company or its stakeholders but do not manage daily operations.
3. **Separation of CEO and Chairperson Roles**
 - Combining the roles of **Chairperson** and **CEO**, as seen in countries like the US and India, concentrates too much power in one individual, leading to potential conflicts in decision-making and governance issues.
 - In countries like the **United Kingdom** and **Australia**, the CEO is prohibited from serving as the Chairperson to maintain a balance of power and ensure accountability. The **CEO** manages the enterprise's operations, while the **Chairperson** leads the board and oversees the evaluation of senior executives, including the CEO, ensuring proper checks on management. Combining the role of both the CEO and the Chairperson removes an important check on senior management's activities.



4. Board Committees (Should the board have committees?)

- Many authorities on corporate governance have recommended the appointment of special committees for
 1. nomination
 2. remuneration (the pay or other financial compensation provided in exchange for an employee's services performed)
 3. auditing (a formal examination of financial records often to uncover fraud or inaccurate tax returns)
- These committees would lessen the burden of the board and enhance its effectiveness.

5. Disclosure and Audit

- There are several issues and questions relating to auditing which have an impact on corporate governance. There are, for instance, questions such as:
 1. Should boards establish an audit committee?
 2. If yes, how should it be composed?
 3. How to ensure the independence of the auditor?
 4. What are the precautions to be taken or what are the positions of the state and the regulators with regard to provision of non-audit services rendered by auditors?
 5. Should individual directors have access to independent resource?
 6. Should boards formalize performance standards?
- These questions are being answered with different perceptions and with different degrees of emphasis.

Governance and Corporate Performance

- **Good corporate governance** ensures effective and efficient company operations in the interest of all stakeholders, It provides assurance that management is acting in the best interest of the corporation, thereby contributing to business prosperity through openness in disclosures and accountability.
- While evidence linking governance to business success is limited, good governance significantly improves the prospect for profitability.
- The key contributions of good corporate governance to a corporation are presented in this topic:
 - Creation and enhancement of a corporation's competitive advantage
 - Enabling a corporation perform efficiently by preventing fraud and malpractices
 - Providing protection to shareholders' interest
 - Enhancing the valuation of an enterprise
 - Ensuring compliance of laws and regulations

Family-Owned Businesses

Strengths

- **Entrepreneurship (Very High):** Highly informal and therefore innovative and flexible.
- **Multiple roles:** A single person at the helm of affairs is often tasked with multiple roles. Therefore, quick decision making is possible.
- **Committed Employees:** Employees are very committed, loyal, shared values and belief system. Family spirit, strong sense of mission/vision.
- **Innovative:** Employees are well spirited and loyal engaged in deeper ties with the family. They can foster creativity, rich interplay of roles and goals.

Weaknesses

- **Complexity:**
 - Family businesses are usually more complex in terms of governance than their counterparts due to the addition of a new variable such as the family.
 - Adding the family emotions and issues to the business increases the complexity of issues that these businesses have to deal with.
 - Unlike in other types of businesses, family members play different roles within their business, which can sometimes lead to a non-alignment of incentives (**something that motivates someone to do something**) among all family members.
- **Informality:**
 - Because most families run their businesses themselves (at least during the first and the second generations), there is usually very little interest in setting clearly articulated business practices and procedures.
 - As the family and its business grow larger, this situation can lead to many inefficiencies and internal conflicts.
 - These inefficiencies and internal conflicts could threaten the continuity of the business.
- **Lack of discipline:**
 - Many family businesses do not pay sufficient attention to key strategic areas such as the CEO and other key management positions, succession planning, family member employment in the company, and attracting and retaining skilled outside managers.
 - Delaying or ignoring such important strategic decisions could lead to business failure in any family business.
 - The failure to address these issues could result in confusion regarding what needs to be done.

- **Lack of Clarity:**
 - Many family-run businesses lack clarity in what is to be done, resulting in confusion.
 - In some cases, they are totally indecisive, resistant to change, and lack management development.
 - There is no organizational chart defining the specific role and responsibility of each family member working for the organization.
- **Autocratic (a style of management in which one leader holds the power to make decisions without input from others):**
 - Sometimes the leadership is as conservative and rigid as a reflection of the autocratic tendency of the family.
 - They resist changing the structure and the system.
 - This resistance arrests the smooth flow of the organization.
- **Family problems:** Can't keep family issues out of business. Inability to balance family's and business's need for liquidity, lack of objectivity, inward-looking, emotionally charged decision-making, can't separate work and family rivalries.
- **Resistance to change:** Family-run business in most cases is tradition-bound. History of family affects business decisions, trust is affected by earlier disappointments.
- **Succession:** In most family-run businesses, the administration finds it difficult to find a suitable successor.
- **Governance:** May favor growth over control, do not have to answer to stockholders. Often no outside board of directors, high premium on privacy.
-

Quiz Insights

Shareholder Trust Improvement: Disclosures and audit practices most likely lead to improved shareholder trust and compliance with regulations.

Lack of discipline: family-run business resistant to hiring external managers for key positions. Overtime, this leads to inefficiencies and internal conflicts.

Topic 13: Components and Facilitators of Corporate Governance

Key Themes:

- Financial Reporting in Corporate Governance
- Whistleblowing Mechanism
- Role of Government in Corporate Governance
- Media Influence on Governance

Keywords:

- Financial Reporting
- Whistleblowing
- Regulatory Role
- Media Influence

Financial Reporting in Corporate Governance

- **Principle IV of the OECD Code:** Emphasizes **Disclosure and Transparency**. This principle says that the corporate governance framework should ensure timely and accurate disclosure on all material matters regarding the corporation like financial situation, performance, ownership and governance. **It provides an elaborate view in relation to this key directive.**
- **A. Disclosure should include, but not be limited to, material information on:**
 1. The financial and operating results of the company.
 2. Company objectives.
 3. Major share ownership and voting rights.
 4. Remuneration policy for members of the board and key executives, and information about board members, including their qualifications, the selection process, other company directorships and whether they are regarded as independent by the board.
 5. Related party transactions.
 6. Foreseeable risk factors.
 7. Issues regarding employees and other stakeholders.
 8. Governance structures and policies, in particular, the content of any corporate governance code or policy and the process by which it is implemented
- **B. Information should be prepared and disclosed in accordance with the high-quality standards of accounting and financial and non-financial disclosure.**
- **C. An annual audit should be conducted by an independent, competent and qualified auditor in order to provide an external and objective assurance to the board and the shareholders that the financial statements fairly represent the financial position and performance of the company in all material respects.**
- **D. External auditors should be accountable to the shareholders and owe a duty to the company to exercise due professional care in the conduct of the audit.**
- **E. Channels for disseminating information should provide for equal, timely and cost-efficient access to relevant information by users.**
- **F. The corporate governance framework should be complemented by an effective approach that addresses and promotes the provision of analysis or advice by analysts, brokers, rating agencies and others, that is relevant to decisions by investors, free from material conflicts of interest that might compromise the integrity of their analysis or advice.**

Engineering Terms:

- **OECD:** Organization for Economic Co-operation and Development.
- **Remuneration:** Compensation or payment.

Whistleblowing Mechanism

Whistleblowing complements the audit activities. They both excel in uncovering different types of cases.

- **Purpose:** Whistleblowing is effective in uncovering a wide range of ethical and behavioral issues, ensuring early detection of unethical practices. It is to be highlighted that whistleblowing must be handled with care to differentiate between true cases and retaliation (to get revenge) or defamation cases. Especially that there is cultural reluctance thereto. (there is a cultural hesitation or unwillingness to accept or engage in whistleblowing.)
- **Audit Activities:** Primarily focus on financial and regulatory compliance matters
- **Whistleblowing mechanism:** ensures early detection or the tendency of organization in pursuing undesirable courses like unethical business practices, murky (dim or gloomy) dealings, nepotism, collusion with independent agencies, political favors, flouting (intentionally not obey a rule) norms of conventions and concepts of accounting

Engineering Terms:

- **Nepotism:** Favoritism granted to relatives or close friends.
- **Collusion:** Secret cooperation for illegal purposes.

- Whistleblowing is defined as an act by an individual who believes that the interest of the public overrides the interests of the organization he or she serves. The act of whistleblowing can have an extraordinary influence on the organization, on the society and on the whistleblower.
- **There are six elements contained in whistleblowing. These are:**
 - The whistleblower
 - The disclosure subject (the act that was perceived)
 - The act of disclosing the wrongdoing
 - The target organization
 - A recipient to whom the disclosure will be made
 - An outcome
- In whistleblowing, the whistleblower has good intentions to prevent wrongdoing and protect the organization.
- The whistleblower never acts negatively to defame the organization or the individuals involved in wrongdoing.
- The whistleblower aims to avoid future wrongdoing that could lead to corporate failure and loss.
- He/she makes sincere efforts within their limits to discourage the wrongdoer.
- The disclosure is made only after these efforts.

Role of Government in Corporate Governance

1. Regulatory Role:

- A large part of the economy of even most of non-centrally planned countries is regulated by the government, as discussed below:
 1. Government may determine the conditions under which persons or corporations may enter certain lines of business as in the granting of a charter, a franchise, a license, or permitting any 'person' to use public facilities or resources.
 2. Government may regulate or assist the conduct of economic ventures of various types once they are under way.
 3. Public control may extend to the results of business operations as in the limitation of public utility profits, ceiling on dividends and imposition of excess-profit taxes on business, etc.
 4. Government may control the relationship between various segments of the economy, the purpose being to settle conflicts of interests or of legal rights and to prevent concentration of economic power in the hands of few monopolies or in a few localities that may cause regional imbalances.
 5. Government may put in place legally constituted regulatory bodies to protect investors, consumers and the general public by ensuring best corporate practices.
 - Government regulation of an economy may be broadly divided into
 - **Direct controls:** For instance, many developing countries have instituted a variety of direct controls over their economies including industrial licensing and price and distribution controls.
 - **Indirect controls:** For instance, a high import duty may discourage imports and **fiscal (of or relating to taxation, public revenues, or public debt)** and monetary **incentives (حافز)** may encourage development of export-oriented industries.

2. Promotional Role:

- In the developing countries, where the infrastructural facilities for development are inadequate and entrepreneurial activities scarce, the promotional role of the government assumes special significance.
- The State will have to assume direct responsibility to build up and strengthen the necessary development of infrastructure such as power, transport, finance, marketing, institutions—for training and guidance and other promotional activities.
- The promotional role of the State also encompasses the provision of various fiscal, monetary and other incentives, including measures to cover certain risks for the development of certain priority sectors and activities.

3. Entrepreneurial Role:

- Public ownership in free societies and their growth in recent times are justified for the following reasons:
 1. In a democracy, the national emergency of war inevitably causes an expansion of State activities, including public ownership, because modern requirements of total war cause people to forsake their convictions concerning private responsibilities and to concentrate on mass power in the State apparatus.
 2. Major economic dislocations, such as the great depression of the 1930s, also tend to stimulate State activities, again leaving a residue of public ownership that takes time and efforts to dissipate, if it is ever finally accomplished.
 3. In economic dislocations, as in the early history of the United States and in the economic development of the developing nations today, government is called on to act as banker, helper or owner of infant industries and generally to expand its central concern for the economy, thus creating considerable degree of public ownership at the outset.
 4. When the private undertakings become unprofitable but the need for their services continues, government may be prevailed upon to acquire and manage such non-profitable business concerns even at a loss.
 5. Governments are also required to extend the owner-manager relationship when there is a pronounced wastage of national resources, or when the threat to them is great, thus diminishing the nation's ability to defend itself or to preserve the bases of a sound economy.

6. Government ownership may also be extended by the failure of private management to consider itself a trustee of the public good and abuse its power, especially in cases where monopoly or semi-monopoly, is then condition.

4. **Planning Role:**

- Governments plan economic growth, resource distribution, and development strategies.

Engineering Terms:

- **Franchise:** Authorization to operate a business.
 - **Fiscal:** Related to government revenue, especially taxes.
-

Stages of News Media Influence on Governance Practices of Corporate:

1. Occurrence of a corporate governance violation by a listed public company in the country.
2. News media communication of the violation to company stakeholders.
3. The actions of company stakeholders in response to news media coverage of the violation.
4. The effects of stakeholder actions on the violating company.
5. The response of the company or corporate governance outcome.

Stakeholder Actions in Response to News Media Coverage of Corporate Governance Violation:

- **Shareholders**
 - Shareholders may require court or governmental protection in case, of corporate governance failure
 - **Policy Makers**
 - Having identified a problem in a given environment, which for the purposes of this discussion is assumed to be the corporate environment, the members of the government formulate a policy through a parliamentary debate, following this, laws and regulations are passed, implemented and evaluated.
 - **The Board of Directors**
 - The board of directors, in its capacity as a monitor of shareholders' interests, has a responsibility to take appropriate action so as to discipline the underperforming or opportunistic management
-

Topic 14 Part B: Ethics in Procurement

Key Themes:

- Understanding ethical principles in procurement.
- Managing ethical risks and challenges.
- Code of Conduct and Stewardship in procurement.
- Transparency, Integrity, and Fairness.
- Conflict of Interest and Ethical Decision-Making.

Keywords:

- **Procurement Ethics** (for general ethical guidelines in purchasing processes).
- **Code of Conduct** (for rules governing behavior in procurement).
- **Conflict of Interest** (for issues of bias in procurement).
- **Transparency** (for open and fair procurement processes).
- **Integrity & Fairness** (for ethical principles in procurement).
- **Supplier Ethics** (for standards applicable to vendors).
- **Fraud & Corruption** (for ethical risks in procurement).
- **Ethical Risks** (for challenges and actions to prevent them).
- **Due Diligence** (for careful evaluation of procurement decisions).

1. Introduction

- The UN procurement process involves a wide range of activities ranging from procurement planning through to contract monitoring and management.
- All procurement practitioners must adhere to the strict ethical standards and responsibilities that apply to procurement activities, in order to protect the integrity, fairness and transparency of the procurement process.
- The discipline of ethics relates to:
 - **Right and wrong conduct** (morally acceptable behavior).
 - **Moral duties and obligations** (responsibilities related to fairness and justice).
 - **System of moral principles and values** (guidelines for ethical behavior).
 - **Moral character** (qualities like honesty and integrity).
- Procurement officials must never use their authority for personal gain and must always act in the **best interest of the UN**.
- Most procurement related principles, i.e., fairness, integrity, transparency and accountability, are based on ethics.

2. Codes of Conduct

- **Definition:** A set of ethical guidelines that outline acceptable behavior.
- Organizations and professions often seek to address standards of conduct through the adoption of codes of conduct.
- Professional codes of conduct are generally written in broad conceptual terms rather than in specific situational or descriptive terms.
- They leave room for interpretation and may often seem ambiguous (doubtful or uncertain).
- Procurement professionals cannot abide merely by the letter of the law, or the specific words in any code of conduct, but should rather be guided by the spirit of the law and the broader concept that the code is intended to express.
- One reason why many procuring organizations avoid detailed and specific codes is that these may give the impression that anything not specifically prohibited is permitted, or that anything not specifically addressed is unimportant.
- No matter how hard policy-makers try, they will never be able to specify in law, code, regulation, rule, or other written requirement everything that a procurement officer needs to know regarding what is allowed or appropriate and what is prohibited or to be avoided.
- It is therefore, necessary for procurement officers to understand what the code, law or rule is intended to accomplish.
- **Key aspects:**
 - Codes are usually broad and open to interpretation.
 - Following only the **letter of the law** (exact wording) is not enough; procurement officers must follow the **spirit of the law** (intended purpose).

3. The Principle of Stewardship

- **Definition:** Stewardship means managing resources responsibly.
- According to the **Concise Oxford English Dictionary**, a steward is “a person employed to manage another’s property.”
- When the steward is a **UN staff member**, spending money that comes from **UN Member States and other public sources** is a **special kind of stewardship**.
- **International procurement officers** are **entrusted** with this responsibility.
- It is an **immense responsibility**, and these funds should **only be spent** in accordance with each **UN organization’s regulations and rules**.

- There should be **no deviation** from these rules to **suit an individual staff member's own convenience**.
- All **UN officials** are required to display the **utmost loyalty** to the **UN, its Charter, regulations, and rules** over their **personal preferences and gains**.
- This requirement is **straightforward and unbending**.
- Exercising the **high standards of professional responsibility** expected of UN staff members is **not always easy**.
- It requires UN staff members to exercise **clear judgment every day** to ensure:
 - **Professional practices** and **outside relationships** conform with the **spirit and letter** of prescribed regulations and rules.
 - Compliance with **broader and harder-to-define ethical standards**.
- **UN procurement officers** face **highly competing pressures**.
- The pressure to **satisfy demands** for:
 - **Quicker turnarounds**.
 - **Better quality**.
 - **Lower prices** in procurement.
- These demands **compete with** the pressure to **fulfill duties** with the **utmost responsibility and ethical standards**.
- The **main role** of each **UN procurement officer** is to:
 - **Execute procurement functions** according to the **highest standards of professionalism**.
 - **Respect the values, objectives, and interests** of the organization.
- This implies that procurement officers must **adhere to the principles of**:
 - **Fairness**.
 - **Impartiality**. (not being biased in one direction or another)
 - **Transparency**.
 - **Stewardship**.
 - **Avoiding conflict of interest** and any **impropriety or perception of such**.
 - **Respecting and applying** the organization's **relevant policies, rules, and procedures**.
- **Key expectations**:
 - Display **loyalty to the UN and its Charter**.
 - Make procurement decisions based on fairness and **avoid conflicts of interest**.
 - Ensure all funds are spent **according to UN rules**.
 - Balance **ethical considerations** with the pressure for lower prices and quick turnarounds.

4. Ethical Concepts and Principles

Loyalty and Respect for Rules

- Professional standards of ethical conduct, no matter what the organization, contain typical characteristics, including commitments to:
 - Loyalty and respect for regulations and rules
 - Integrity
 - Impartiality, fairness and transparency
 - Confidentiality
 - Avoidance of appearance of impropriety
 - Due diligence (**the steps an organization takes to thoroughly investigate and verify an entity before initiating a business arrangement**)

Integrity

- **Definition**: Maintaining strong moral principles like honesty and incorruptibility.
- **UN procurement officers** are expected to maintain **superior standards of integrity and moral values**.
- The **International Civil Service Advisory Board** identifies **integrity** as:
 - **"One of the fundamental, if not paramount, standards of conduct."**
 - **Which is, "Underlined in Article 101 of the United Nations Charter and explicit or implicit in corresponding articles of the basic instruments of the specialized agencies."**
- Their report explains that **integrity**, as enshrined in the **UN Charter**, includes:
 - "The concept of integrity enshrined in the Charter of the United Nations embraces all aspects of behaviour of an international civil servant, including such qualities as **honesty, truthfulness, impartiality, and incorruptibility**. These qualities are as basic as those of competence and efficiency, also enshrined in the Charter..."
 - These qualities are **as essential as competence and efficiency**.
- **Cultural differences** (nationality, ethnicity, industry, or profession) must be **set aside**.
- **Generic principles of integrity that extend beyond and rise above such differences** must be **allowed to prevail**, especially in connection with the **business transactions conducted by UN procurement officers**.
- **Integrity, to a procurement officer in the international marketplace**, means believing that **public trust is so important that it cannot be compromised**.

- A **procurement officer** should demonstrate integrity by:
 - **Making decisions without consideration for personal gain.**
 - **Resisting political pressure** in decision-making.
 - **Ensuring that official power or authority are not abused.**
 - **Demonstrating UN values in his daily actions**, including:
 - **Impartiality.**
 - **Fairness.**
 - **Honesty.**
 - **Taking prompt and effective action** to address **unprofessional or unethical behavior.**

Impartiality and Fairness

- **Impartiality:** Treating all suppliers equally.
- In a report by the **International Civil Service Advisory Board**, **impartiality** is identified as a **key requirement**.
- The report states that **impartiality** includes:
 - “Impartiality implies **objectivity, lack of bias, tolerance, restraint** - particularly when **political or religious disputes or differences arise**. The staff member’s **personal views and convictions remain inviolate**, but he has not the freedom of a private person to “take sides,” to enter a dispute as a partisan, or publicly to express his convictions on matters of a controversial nature, either singly or as a member of a group. Just as the practice of impartiality will strengthen the secretariat, repeated instances of partiality, or bias, will do serious harm to the organization”.
 - **Objectivity.**
 - **Lack of bias.**
 - **Tolerance and restraint(limit)**, especially in **political or religious disputes**.
 - **Maintaining personal views privately** (staff members cannot publicly express opinions on controversial issues).
 - **Avoiding taking sides** in disputes.
 - **Strengthening the secretariat** through impartiality, while **partiality or bias** can seriously harm the organization.
- In the context of impartiality and fairness the procurement officer should:
 - **Set aside all personal and organizational biases.**
 - **Apply the same standards to all suppliers** (equal treatment).
 - **Examples:**
 - If one supplier requests **additional information**, all suppliers must receive the **same information at the same time**.
 - If one **offer is disqualified in the evaluation process** and the award placed with the next lowest priced offer, the reason for disqualifying the lowest offer must be applied to the evaluation **of all offers uniformly**.
 - **Fairness** implies:
 - Being **reasonable and impartial**.
 - Treating **UN suppliers professionally** and with **business-like courtesy (politeness/ respect)**.
 - **Strictly following policies and procedures** for conducting the transaction

Transparency

- **Definition:** Ensuring openness in procurement activities.
- **Transparency** means **unimpeded visibility (clear and unrestricted view)**.
- Because **public procurement** involves the **use and accountability of public funds**, transparency is **paramount** (more important than anything else) in procurement activities.
- **All transactions** are subject to **scrutiny (examination)**, but the **UN faces a higher level of scrutiny** compared to other organizations.
- **Therefore, Procurement officers** must always conduct themselves in a way that ensures:
 - **Scrutiny does not damage the UN**, its **leaders, member organizations, staff, or programs**.
- **Two Degrees of Transparency**
 - **Internal Scrutiny** (within the UN):
 - **Such as Examinations conducted by internal auditors.**
 - Seeks a certain **compliance with UN’s own standards by the UN’s own compliance officers**.
 - It enables an organization to determine if
 - There are **problems related to transparency that need to be addressed**.
 - Provides **recommendations for improvement** on what steps may need to be taken.
 - **External Scrutiny** (outside the UN):
 - **Examination by external bodies** such as:
 - **Member States.**
 - **The press.**
 - **External auditors.**
 - **Other outside observers.**
 - It refers to the notion that almost anyone can observe **UN activities** and see how the **UN conducts its business**.

Confidentiality

- **Definition:** Protecting sensitive procurement information.
- **Confidentiality** is especially important in **UN procurement** due to the **sensitive nature** of the information that is handled in procurement processes.
- Procurement processes handle **delicate data**, such as:
 - **Pricing of products.**
 - **Marketing strategies.**
- A **breach of confidentiality** can lead to:
 - **Discrediting the UN.**
 - **Instilling distrust (loss of trust)** from **governments, partners, and suppliers.**
- Confidentiality might seem to be in direct contradiction to the concept of transparency but, in this context, it means that whilst the way the overall procurement process is conducted needs to be clear and transparent, true proprietary data needs to remain confidential.

Avoiding the Appearance of Impropriety

- **Definition:** Even if actions are technically legal, they must also appear ethical to observers.
- In the **private sector**, appearances are influenced by the **wishes or standards** of the organization's leadership.
- Employee behavior is judged based on:
 - **Legality.**
 - **Satisfaction of owners or customers.**
- **However, International public-sector procurement officers** must adhere (follow) **more conservative standards.**
- **UN procurement officers** must:
 - Be constantly aware of **how their actions appear to outside observers.**
 - Always behave in such a way that their actions **cannot be perceived as improper.**
- **Behaving correctly in a "technical" sense is not enough;** it is also necessary to **avoid even the appearance of impropriety (unacceptable).**
- **Not all societies have the same standards or traditions** when determining **what is considered proper and what is not.**
- It is important to recognize **cultural differences** in **appearances** and to anticipate, **in the most conservative terms**, what might be **perceived as improper conduct.**
- This added dimension of the appearance of impropriety places an **extra responsibility on procurement professionals.**
- Procurement officers must:
 - **Recognize cultural differences** in appearances.
 - **Anticipate, in the most conservative terms**, what might be perceived as improper conduct.
- **What people think of the procurement officer's behaviour** can:
 - **be the basis of major scandals**, based on misunderstandings and erroneous information that can damage an organization's effectiveness in achieving its mission.
 - **In an international environment**, the damage can be **immense and have long-lasting effects on the credibility of the organization.**

Due Diligence

- **Definition:** Careful and thorough performance of duties.
- **Due diligence**, in the context of UN procurement, refers to **carrying out duties carefully and thoroughly** and avoiding careless practices or techniques. Due diligence requires that all activities by procurement officers be **pursued in a manner that goes beyond the minimum effort.** For example, diligent UN procurement officers should:
 - Check the references of potential suppliers.
 - Develop impartial evaluation criteria.
 - Carefully analyse the offers received.
 - Avoid cutting corners for the sake of convenience.

5. Ethical Risks and Actions to Manage Them

Conflict of Interest

- **Definition:** A situation where personal interests interfere with professional duties.
- A very common risk situation related to **ethics** in **procurement** is the risk of a **conflict of interest.**
- A **conflict of interest** can be defined as:
 - A direct or mutually exclusive (*incompatible*) clash between the interest of the **UN** and the **private or personal interest** of a UN procurement officer.
 - Occurs when a staff member's **private interests**, such as **outside professional relationships** or **personal financial assets**, interfere or appear to interfere with the proper performance of their **professional functions** or **obligations** as a UN official.
- Within the procurement environment, a **conflict of interest** may arise in connection with:
 - **Private interests**, such as **personal investments and assets.**
 - **Political or other outside activities.**

- **Employment** after **retirement** from the UN.
- Receipt of a **gift** that may place the staff member in a position of **obligation** (*responsibility*).
- A **conflict of interest** may also arise in situations where:
 - A staff member is seen to **benefit**, directly or indirectly, from their **decisions**.
 - A **third party**, including **family, friends**, or someone they **favor** (*prefer*), **benefits** from the staff member's **decisions**.
- **Types of conflict of interest** in the procurement environment:
 - **Actual Conflict of Interest** - When a staff member is in a position to be **influenced** by their **private interests** while performing their **job**.
 - **Perceived Conflict of Interest** - When a staff member is in a position to **appear** to be **influenced** by their **private interests** while performing their **job**.
 - **Potential Conflict of Interest** - When a staff member is in a position where they **may be influenced** in the **future** by their **private interests** while performing their **job**.
- In the context of **procurement**, a UN procurement officer should:
 - **Declare with immediate effect** any **potential conflict of interest**.
 - **Not use information** obtained during their **professional duty** for **personal profit** (*gain*).
 - **Disclose** and **dispose** the **financial interest** involved.
 - **Withdraw** from any **procurement process** where they may have a **conflicting interest**.

Declaration

- It is **good practice** to have officials involved in the **procurement process**, including those participating in:
 - **Bid opening panels**.
 - **Evaluation committees**.
 - **Contracts committees**.
- Officials should **sign**, in advance of their duties, a **declaration of no conflict of interest**.
- **Honesty, truthfulness, impartiality** (*fairness*), and **incorruptibility** (*integrity*) must always be applied whenever:
 - A **conflict of interest**, or the **appearance of conflict of interest**, arises during the **procurement process**.

Gifts and Gratuities

- **Common issue**: Suppliers may offer gifts to influence procurement decisions.
- A common **conflict of interest** situation UN **procurement officers** face is whether or not to **accept gifts** from:
 - **Suppliers**.
 - **Partners**.
 - **Governments**.
- Offering **gifts** to customers is a common practice in the **private sector**. It is a **marketing strategy** based on the universal sense of **reciprocity** (*mutual exchange*):
 - If we **receive** something, we feel **obliged to give something in return** (i.e., "there is no such thing as a free lunch").
- Suppliers often offer various types of **gifts**, including:
 - **Perishable** (*likely to die or decay*) **products**.
 - **Hospitality**.
 - **Free training courses** or **experiences**, such as:
 - **Exhibitions**.
 - **Fair trades**.
 - **In-kind donations** (*non-monetary contributions*) related to the activity of the specific UN organization.
- **Identifying covert gifts** (*hidden or subtle offerings*) is not always easy:
 - A supplier may offer **free training activities**, which might appear **beneficial** for the organization.
 - A careful **review** should assess the **impact** of accepting such gifts:
 - Would receiving the gift **benefit one company** over others?
 - Would acceptance be **fair** to competitors?
- It can be difficult to judge:
 - How to **behave** in a particular situation,
 - What the **appropriate action** would be.
- Accepting **gifts, honors**, or other **tokens of appreciation** can:
 - Impact a **procurement officer's independence** and **impartiality** (*fairness*).
 - Compromise their **integrity** and the UN's **reputation**.
 - Create a **conflict of loyalties** or raise **expectations** from the giver of the gift.

6. Potential Areas of Risk in the Procurement Cycle

- **Procurement officers** are exposed to a number of **risks** when dealing with **suppliers** whose **business practices** may differ from those of the **UN organization**.
- **Situations posing Ethical dilemmas** can occur in any type of **procurement activity**, but they have a **higher impact** in:
 - **High-value purchases.**
 - **High-risk purchases.**
- Regardless of the **value** of the procurement activity, the **highest standards** of **ethical behavior** must always be maintained to ensure the **probity** (*integrity*) of the procurement process.
- Some **potential areas of risk** related to **ethics** in the **procurement cycle** are listed in the **table below**.

Table 1 – Potential areas of risk

AREA OF RISK	EXPLANATION AND EXAMPLES
Budgeting	Lack of proper budgetary control, for example when all funds are not allocated to a specific purpose, can provide the necessary funds for fraud and corruption to take place.
Financial approval policy	Without regular audits and strict management controls, there are opportunities for fraudulent use of a person's own levels of authority, or of misusing someone else's.
Perceived need	Requirements can be invented or falsified.
Specification development	• Specifications can be written to favour a specific supplier. • Clarifications on specifications can be provided to only one of the invitees during the tendering process.
Evaluation criteria	Evaluation criteria can be written, or amended, after receipt of offers to favour a particular supplier.
Pre-qualification	This process can be used to limit the field of competition to give a favoured supplier an advantage.
Invitation to tender/sourcing	This process can be used to give the illusion of competition where it does not really exist by inviting tenders from companies who are known to be unsatisfactory or ineligible, or by not sending out complete specifications to all bidders at the same time etc.
Offer evaluation	Fraud at this stage occurs mainly when objective evaluation criteria have not been agreed in advance. It can also occur where technical staff are able to use their specialist knowledge to mislead other members of the evaluation team.
Negotiation	Favoured suppliers can be assisted or given useful information during negotiations.
Contract award	Suppliers may find opportunities for fraud either through deliberately fraudulent acts or through buyer incompetence. Procurement staff could artificially split contract award to remain under the organizational threshold for contract committee review or to award more contracts to preferred suppliers.
Post award changes to specifications	Allow suppliers to increase profits by charging more for revised specifications.
Ordering and receipt of goods	Examples are: • Allowing under-deliveries of goods or non-performance to specifications • Drafting false goods inward notes • Deliberately over ordering • Allowing inventory to dwindle so that emergency orders at a higher price will have to be processed
Invoice certification	Deliberate overcharging, backdating orders to allow benefit from price changes, paying twice, failing to insist on or monitor retentions.
Decentralized procurement organization	Delegations to a large number of people making it difficult for the central procurement function to exercise total control and awareness of what is going on in the decentralized units. Although decentralization can often improve efficiency and reduce costs, it can also increase the risks of corruption.

- **Examples of risks:**
 - **Budgeting:** Poor control can lead to fraud.
 - **Specification Development:** Favoring one supplier over others.
 - **Evaluation Criteria:** Changing criteria after offers are received.
 - **Negotiation:** Giving advantages to preferred suppliers.
 - **Post-Award Changes:** Increasing supplier profits unfairly.
 - **Receiving Goods:** Accepting lower-quality products.
 - **Invoice Certification:** Overcharging or backdating invoices.

Potential warning signs of unethical practices

- There are some typical signs that may indicate or warn of unethical practices. These include, but are not limited to the following:
 - Deviation from **proper procedures**
 - **Poor record keeping**: missing files and documents
 - **Buyer's extravagant (expensive) life style**
 - Resistance to **audit and** reluctance(unwillingness) to **delegate**
 - Exclusive handling of supplier **by only one person**
 - **Secrecy Vs confidentiality**
 - **Dictatorial management style**
 - **Unnecessary** meetings with suppliers
 - **Reluctance of suppliers** to submit bids

Below is a **six-step ethical decision-making model** which UN staff should consider when faced with making ethical decisions.:

1. **Evaluate the situation** (determine the facts that you know and check your perceptions).
 2. **Test the ethics** (What are the ethical issues and questions at stake?).
 3. **Help those affected** (Who are the stakeholders, and how might they be affected by the situation?).
 4. **Identify the rule(s)** (Ensure compliance with organizational codes of conduct).
 5. **Call for advice and guidance, if necessary**
 6. **Stand up for ethics.** (Make the best possible decision and take action. Ensure the resolution is just and fair for all stakeholders).
- In addition to the **six-step process** outlined above, **UN staff** should:
 - Ensure the decision is **consistent** with their organization's:
 - **Policies.**
 - **Procedures.**
 - **Guidelines.**
 - Confirm it is **acceptable** under applicable **laws** and **regulations**.
 - Verify that the decision adheres to the **universal principles** and **values** adopted by their organization.
 - Ensure it satisfies the staff member's definition of:
 - **Right.**
 - **Fair.**
 - **Transparent.**
 - **Impartial** (*unbiased*).

7. Ethics at an Organizational Level

There are also warning signs at the organization level showing potential weaknesses in ethical practices.

Some of these are included below:

- **No communication** of expectations.
- Too much **trust** in key employees.
- **Lack of proper authorization procedures** -- too many **exceptions**.
- **Changes** in organizational **structure**.
- A tendency towards **crisis management**.
- **Poor/no system design**—processes just “**evolved**” over time.
- **Accounting personnel** who are inexperienced or **lax** (*careless*) in their duties.

Organizational Tools and Mechanisms to Prevent and Detect Unethical Practices

In addition to individual UN staff taking responsibility for their ethical behaviour and following best practices, there are also some **organizational tools and mechanisms** that may be used to **prevent and detect the occurrence of unethical practices** as outlined below.

Management Responsibility

- Management should:
 - Maintain the **highest standards of integrity** in its everyday dealings.
 - Management's responsibility is to **set the highest standards of integrity** and **to be an example for everybody in the organization to follow**.
 - Managers should also **point out correct behaviour** to employees and draw the line between **acceptable and unacceptable behaviour**.
- **Senior management** dishonesty can lead to the spread of **corruption** and **fraud** at all levels.

- Management is also ultimately responsible for:
 - The organization's **operations** and **assets** under their command.
 - It is also their responsibility, and in their interest, to ensure that the organization has the necessary **procedures** and **control systems** to ensure:
 - Maximize **security**.
 - Minimize the risk of **corruption** and **fraud**.

Code of Ethics

- It is critical for all UN organizations to:
 - Clearly document the expected **behaviour** from staff members as well as what is considered **unacceptable behaviour** in a **code of conduct** or **code of ethics**.
- It is also important that organizations:
 - **Promulgate** (*publicize*) and **enforce** their codes of ethics.
 - So that they promote an atmosphere of **ethical behaviour**.

Organizational Procedures

- To **prevent fraud and corruption** an organization should have in place the following organizational procedures:

PROCEDURES	INCLUDING
Pre-employment screening	Background of all job applicants should be checked before they are employed and granted access to premises and assets.
Classification and protection of information	'Clear desk policy', secure filing cabinets for all employees, sufficient number of paper shredders, secure disposal of all waste paper.
Data security standards	• Procedures should be introduced for all data processing resource. • Instructions on minimum standards should be enforced.
Incident reporting	All employees should be responsible for reporting losses and security incidents. All incidents, regardless of how small they are, should be reported.

Personnel policies and procedures

Fair, open, and efficient **personnel policies** and **procedures** reduce the organization's **exposure to fraud**. UN organizations should consider the following factors in **their personnel policies and procedures**:

POLICIES AND PROCEDURES	FACTORS TO CONSIDER
Job descriptions	Security responsibilities should be drafted into contracts and job descriptions.
Education and training	• Awareness training can clarify what is meant by ethical conduct, short cuts, and fraud. • Training programs can be supported by booklets on the organization's business ethics and security policies, newsletters including reports on frauds discovered and lessons learned from them, as well as films and videos. • New employee induction training should cover security.
Investigation	• Organizational rules for conducting investigations into suspected or reported incidents of fraud or breaches of security, including employee obligations to assist in such investigations. • Criminal offences or reasons for disciplinary action should be brought to the attention of all staff.

Accounting Controls

- The **integrity** of accounting systems is essential in preventing **fraud**.
- Controls should:
 - Ensure that details of all goods and equipment moving in and out of the organization are recorded on **serially numbered documents** or **computer records**.
 - Retain copies of documents recording movements **securely**.
 - Make maximum use of **numerical controls**.
- **Levels of authority** to approve accounting transactions should:
 - Be clearly defined and regularly **audited**.
- Each system should:
 - Define who is responsible for **losses**, **errors**, and **concealment** (hiding something).
 - Identify and enforce **areas of responsibility**.
- **Books and records** should be protected like all other **organizational assets**.

Segregation of Duties

- The most basic and effective way to prevent **fraud** since it removes the possibility of '**closed loops**'.
 - **Closed loops**: one person having the authority to budget, provision, order and pay.
 - Each transaction should be divided into a number of stages and no one person should ever have the authority to handle all of the stages.

- All **purchasing responsibilities** (e.g., supplier selection) must be segregated (separated) from **disbursement** (the delivery of money from a fund) and **accounting activities** to ensure **fairness** and **transparency**.

Control Systems

- **Control systems** in procurement:
 - Protect honest buyers and suppliers from false accusations.
 - Encourage **honest and effective** work.
 - Detect and prevent **corruption**.
- It is often difficult to get the right balance between **under and over control** and many organizations have control systems which are either **too restrictive or too lax**.
- An effective control system must:
 - Be **well-balanced**—neither too restrictive nor too lax.
 - Be **flexible**, allowing honest buyers and suppliers to operate efficiently while at the same time minimizing risks of dishonesty.

Controls in Procurement

- Procurement procedures should be set-out in a **manual** for all staff involved in procurement.
- **Procedures, authorities, responsibilities and penalties** for not adhering to procedures should be clearly defined.
- Considerations when setting procurement strategies:
 - Clearly define and enforce the **threshold** for competitive tenders. (Threshold above which contracts and orders must be put out or competitive tenders should be clearly defined and enforced.)
 - Avoid **spot, short-term, or emergency orders** whenever possible. (As far as possible, spot, short-term, or emergency orders should be avoided.)
 - Minimize the use of **cost-plus contracts** and carefully verify the supplier's expenses when unavoidable. (Cost-plus contracts should be avoided if possible, but if they cannot be avoided special care should be taken to verify the supplier's expenses.)

Control Over Goods and Services Received

- As discussed above, there are several potential risks related to ethics in the procurement cycle. Controlling the receipt and storage of goods is a vital aspect of **eliminating opportunities for fraud**.
- Measures to avoid risks, The UN organization should:
 - Have an **auditable system** for receiving and verifying goods' **quality and quantity**.
 - There should be procedures in place for **reporting and recording** variations in the **quantities** of goods and any **defects** in quality such as **damage or breakages**.
 - **In the case of large quantities**, the organization should implement a **sampling system** in order to check the accuracy of the supplier's shipment.
 - UN staff responsible for receiving goods should be properly trained in inspecting the quality of materials received.
 - In addition, only goods/services supported with an **authorized purchase order or contract** can be accepted.
 - The UN organization must have a system in place to verify that goods or services received are for legitimate outstanding purchase orders
 - This entails **matching the purchase order number** entered as part of the receiving transaction with records contained in the organization's procurement system.
- Organizations should:
 - Periodically and systematically monitor their suppliers to ensure that their actual performance meets expectations.
 - Performance reporting includes:
 - Percentage of **on-time delivery**
 - Shipment accuracy
 - Product quality
 - Actual cost of performance compared with original cost projections.
- Any storage areas maintained by a UN organization should:
 - Be secure and accessible only to authorized personnel who have been given special access cards.
 - The organization should also have **electronic log** of stored and removed items from the storage facility.
 - **Spot checks** should be carried out **periodically**.
 - There should be a UN designated staff member **responsible for monitoring the quantity of stocks of frequently used items** in order to ensure that they are purchased on a **regular basis** and that **last-minute purchases will not be required**.

8. Ethics at the Supplier Level

Duty to Report

- If you **see something, say something**.
- Management should maintain an **open-door, speak-up culture** where staff and other personnel can approach them with concerns of any nature, **without fear that they will be punished for doing so**.
- If you see, or suspect, that a UN staff member, or other personnel, or a vendor, supplier, partner, government official, NGO or anyone else is acting or has acted improperly (or if you receive credible information from someone else that leads you to believe there may be

improper activity taking place, such as harassment, fraud or corruption etc.), you have an obligation to report the suspected misconduct without fear of reprisal (punish them because they have done something unpleasant to you) or retribution (punishment which is carried out by someone other than the official authorities).

- **Reporting misconduct** (e.g., harassment, fraud, corruption) is an obligation when:
 - A UN staff member, personnel, or a vendor acts improperly.
 - Credible information suggests improper activity.
- Reporting channels:
 - Speak to your manager or, if you have concerns raising the matter with your manager, speak to another manager.
 - Alternatively, use the **Office of Audit and Investigations Hotline** for anonymity (except in cases of workplace harassment or abuse of authority).
 - the **Office of Audit and Investigations** maintains an Investigations Hotline for that purpose.
 - personnel who report wrongdoing have the option of **remaining anonymous** (except in matters alleging workplace harassment or abuse of authority because the victim must identify himself).
 - Finally, most UN organizations have established an **Ethics Office** which can be approached for confidential advice and guidance. UN staff members should familiarise themselves with their own organizations policies and procedures governing ethics in procurement.

Flipboards Notes: (Not clear)

Learning Organization (Peter Senge, "The Fifth Discipline")

- **Definition:** A learning organization is one that continuously transforms itself by enabling its members to grow and adapt to achieve their goals.
- **Key Disciplines:**
 - **Personal Mastery:** Commitment to lifelong learning.
 - **Mental Models:** Challenging and improving assumptions and mindsets.
 - **Shared Vision:** Aligning team and organizational goals.
 - **Team Learning:** Collaboration and collective growth.
 - **Systems Thinking:** Understanding interdependencies and the "big picture."

CFO (Chief Financial Officer)

- **Role:** Senior executive responsible for managing financial actions and strategy.
- **Key Responsibilities:**
 - **Forecasting and Planning:** Predicting financial trends and budgeting.
 - **Accounting:** Ensuring accurate financial reporting and records.
 - **Compliance:** Adhering to financial laws, regulations, and ISO standards.
 - **Business Controls:** Establishing systems to manage risks and monitor performance.

Transactions in Business

- Involves the exchange of goods, services, or funds between parties.
- Types include:
 - **Sales Transactions:** Purchase of goods/services.
 - **Financial Transactions:** Loans, investments, or payments.
 - **Operational Transactions:** Inventory or supply chain processes.

Forecasting

- Predicting future business trends using historical data and market analysis.
- Helps in:
 - Strategic planning.
 - Resource allocation.
 - Risk management.

Backing Up Estimation and Assumptions

- Use of data and analysis to validate assumptions.
- **Confidence Levels:** Indicate the reliability of the predictions (e.g., 90%, 70%).

CPA (Certified Public Accountant) and RCA (Root Cause Analysis)

- **CPA:** A licensed professional in accounting, auditing, and taxation.

- **RCA:** A method used to identify the underlying cause of a problem to prevent recurrence.
-

Consulting Process

- **6 Value-Generating Activities:**
 - Identifying opportunities.
 - Problem-solving.
 - Strategic alignment.
 - Implementation of solutions.
 - Performance enhancement.
 - Sustainable value creation.
 - **4 Value-Enabling Activities:**
 - Research and analysis.
 - Stakeholder engagement.
 - Resource allocation.
 - Monitoring and evaluation.
-

RAID (Risks, Assumptions, Issues, Dependencies)

- A project management framework for tracking:
 - **Risks:** Potential problems.
 - **Assumptions:** Factors assumed to be true.
 - **Issues:** Current challenges.
 - **Dependencies:** Interlinked tasks or resources.
-

BANT (Budget, Authority, Need, Timeline)

- A qualification framework for sales leads:
 - **Budget:** Can the buyer afford the solution?
 - **Authority:** Does the buyer have decision-making power?
 - **Need:** Does the buyer require the solution?
 - **Timeline:** When will the buyer make a decision?
-

Conditions of Satisfaction

- Clear criteria for success agreed upon by all stakeholders.
-

30% CS (Customer Satisfaction)

- A metric reflecting customer satisfaction with a product/service.
-

Learning, Stalling, Service Quality

- Continuous learning improves service quality, while stalling hinders growth.
-

Causality Chart

- Visual representation of cause-and-effect relationships to analyze problems.
-

L&D -> Knowledge Certified -> 30% CS

- **Learning and Development (L&D):** Focused on upskilling employees.
 - Certified knowledge leads to improved **customer satisfaction (30%)**.
-

HC -> Quality Assurance -> 30% CS

- **Human Capital (HC)** initiatives ensure quality assurance and improved customer satisfaction.
-

Opened -> Collaboration (OPE, RCE)

- Encouraging collaboration through **Operational Process Excellence (OPE)** and **Resource Collaboration Efficiency (RCE)**.
-

Protecting Information

- **Types of Information:**
 - Competitive information.
 - Sensitive data.
 - Intellectual capital (specific knowledge assets).
-

KSA (Knowledge, Skill, Attitude) -> Manpower

- Core competencies required for workforce productivity:
 - **Knowledge:** Theoretical understanding.
 - **Skill:** Practical abilities.
 - **Attitude:** Behavioral disposition.
-

Capacity

- The organization's ability to handle workloads and achieve objectives.
-

Objective Data -> Subjective Decision-Making

- Using measurable data to guide intuition-based decisions.
-

Secondary Controls -> 80/20 Pareto Selection

- **Secondary Controls:** Backup or contingency measures.
 - **80/20 Rule (Pareto Principle):** Focus on the 20% of actions yielding 80% of results.
-

COTS (Commercial Off-The-Shelf)

- Software solutions:
 - **SAP** works with **ERP** (Enterprise Resource Planning).
 - **Salesforce** and **Siebel** work with **CRM** (Customer Relationship Management).
-

DDD (Domain-Driven Design)

- Provides a **middle ground** for aligning business goals and software solutions.
-

App (Teko)

- Used for billing and transaction management.